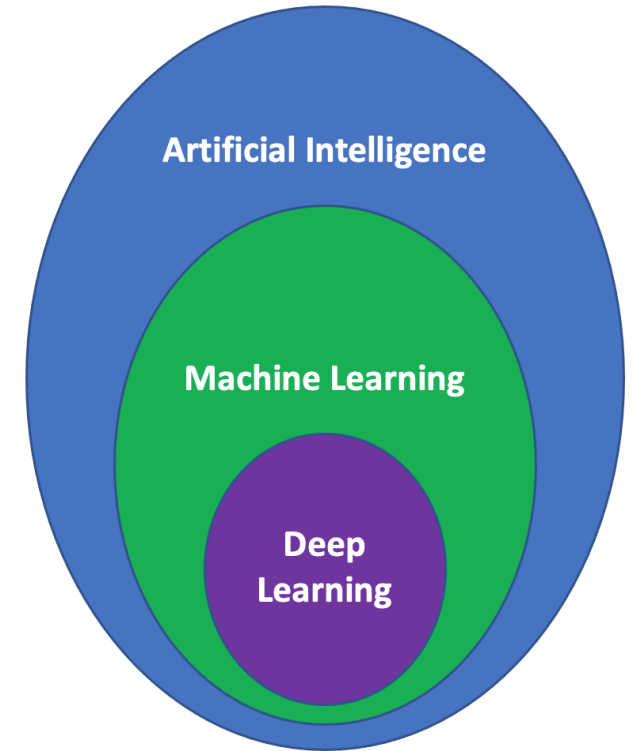


Deep Learning

Lecture 1

What is Deep Learning

- **Artificial Intelligence (AI)** any technique that enables computers to mimic human intelligence (e.g., perception, comprehension, attention, problem solving, planning) from multiple modalities
- **Machine Learning (ML)** is a subset of AI that learn through use of data
- **Deep Learning (DL)** is a subset of ML that involves many parameters and large amounts of data



First use-case: Image classification

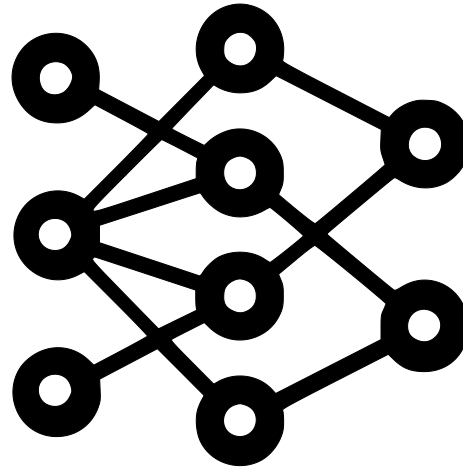
Real world input



Model
input

$$\begin{bmatrix} 124 \\ 140 \\ 156 \\ 128 \\ 142 \\ 157 \\ \vdots \end{bmatrix}$$

Model



Model
output

$$\begin{bmatrix} 0.00 \\ 0.00 \\ 0.01 \\ 0.89 \\ 0.05 \\ 0.00 \\ \vdots \\ 0.01 \end{bmatrix}$$

Real world output

Aardvark
Apple
Bee
Bicycle
Bridge
Clown
⋮

Followed by other tasks: Image segmentation

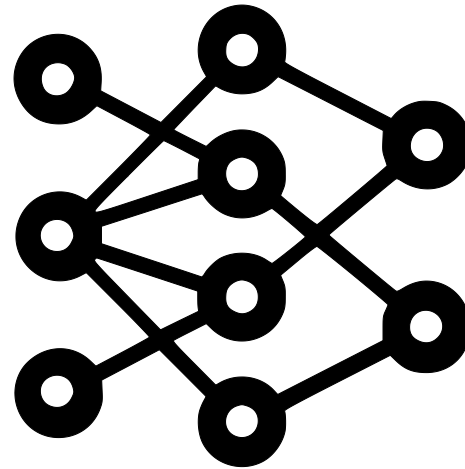
Real world input



Model input

$$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 185 \\ 204 \\ 232 \\ \vdots \end{bmatrix}$$

Model



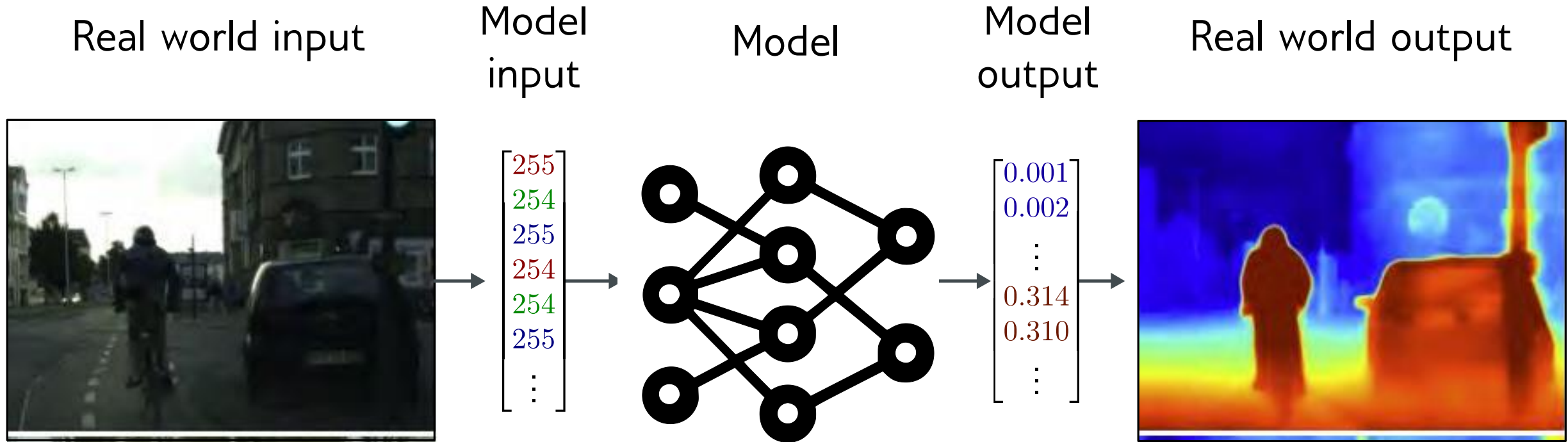
Model output

$$\begin{bmatrix} 0 \\ 0 \\ 0 \\ \vdots \\ 1 \\ \vdots \end{bmatrix}$$

Real world output

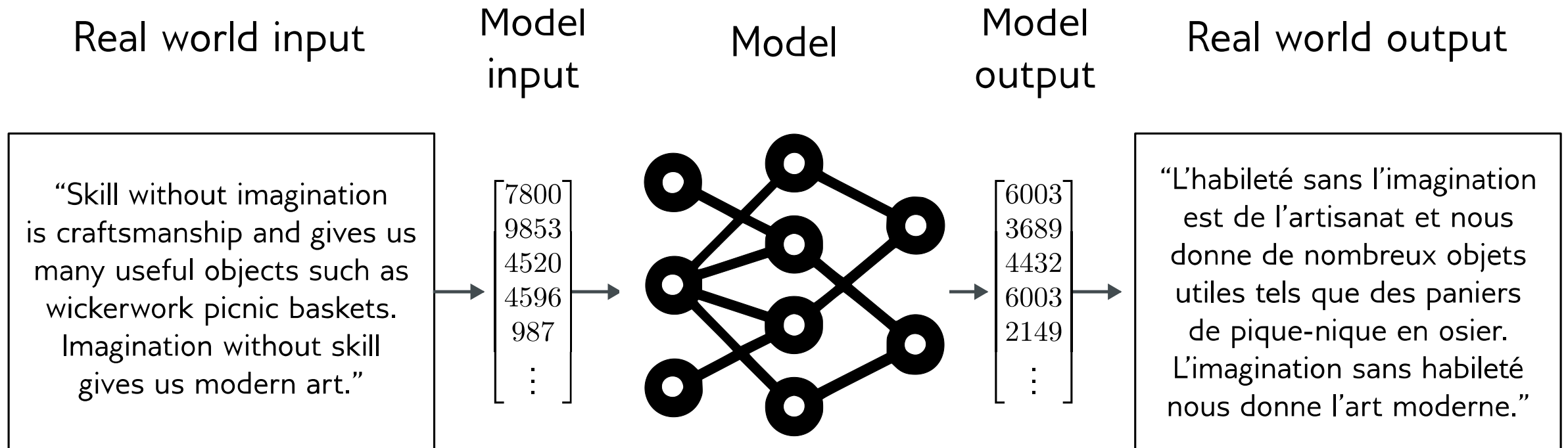


Depth estimation etc



(Lead big rise in autonomous driving cars)

Next, language processing, e.g., Translation



Multimodal: Image captioning

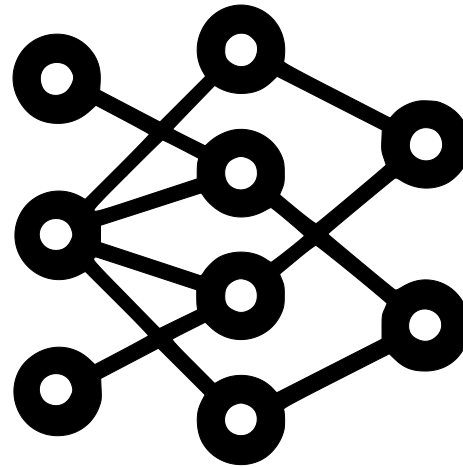
Real world input



Model
input

$$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 185 \\ 204 \\ 232 \\ \vdots \end{bmatrix}$$

Model



Model
output

$$\begin{bmatrix} 1 \\ 5593 \\ 7532 \\ 7924 \\ 1 \\ \vdots \end{bmatrix}$$

Real world output

“A Kazakh man on a
horse holding a
bird of prey”

Followed by Visual Questions

Real world input

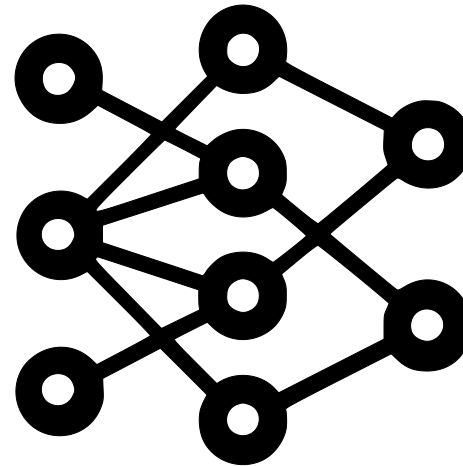


What is unusual?

Model
input

$\begin{bmatrix} 183 \\ 204 \\ 231 \\ 4596 \\ 987 \\ \vdots \end{bmatrix}$

Model



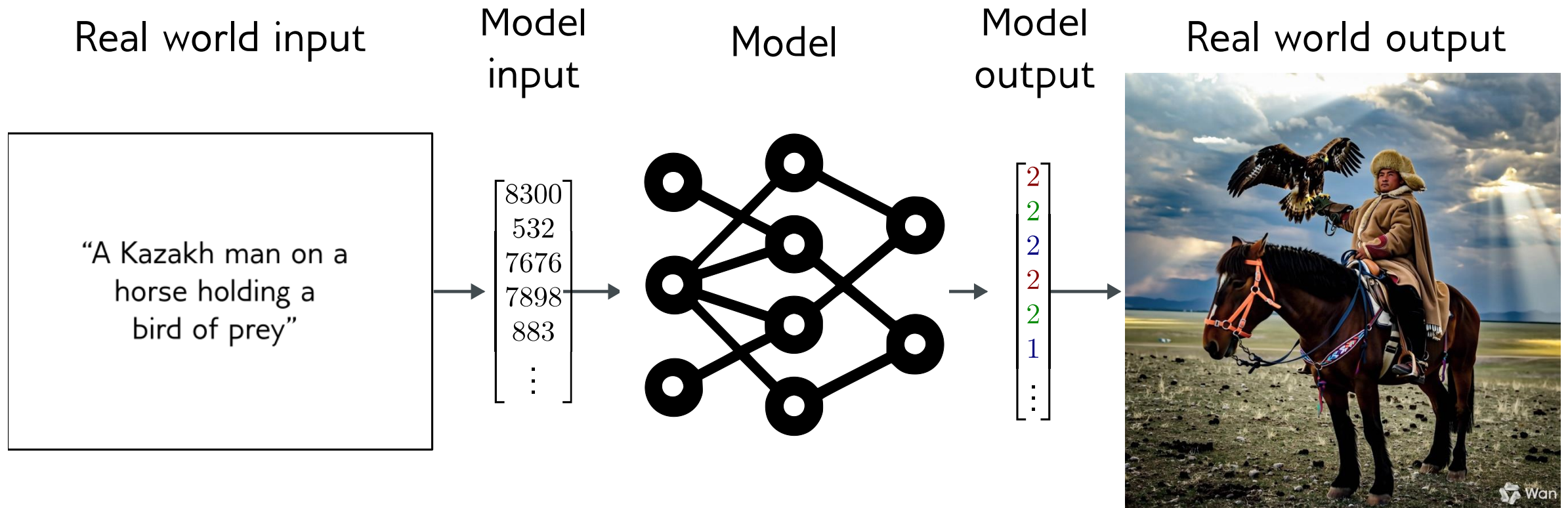
Model
output

$\begin{bmatrix} 6003 \\ 3689 \\ 4432 \\ 6003 \\ 2149 \\ \vdots \end{bmatrix}$

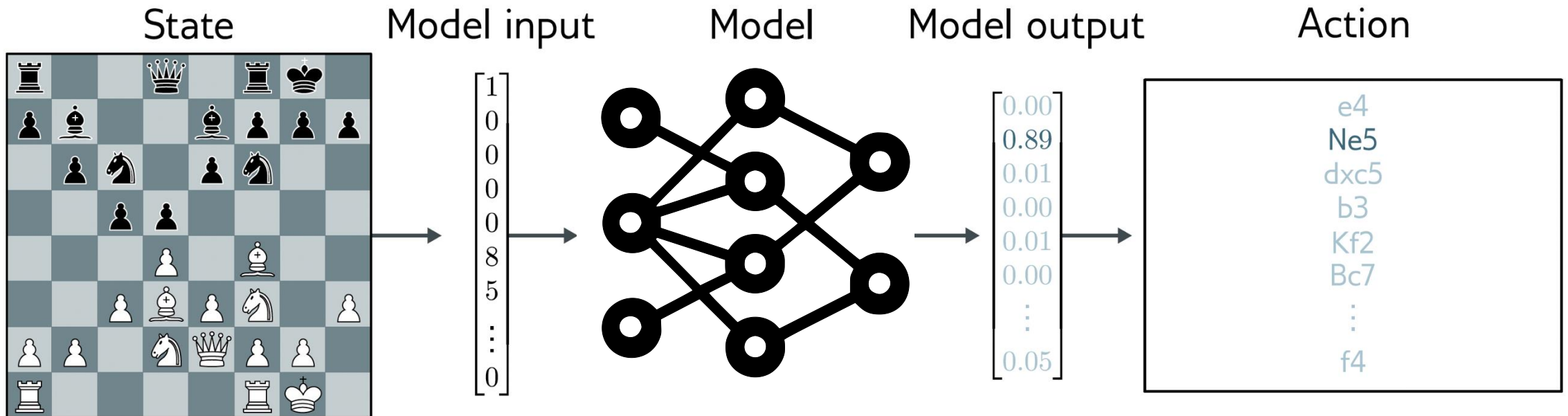
Real world output

The unusual thing about this image is that a man is ironing clothes on an ironing board attached to the roof of a moving taxi

The generative era: Text-to-Image



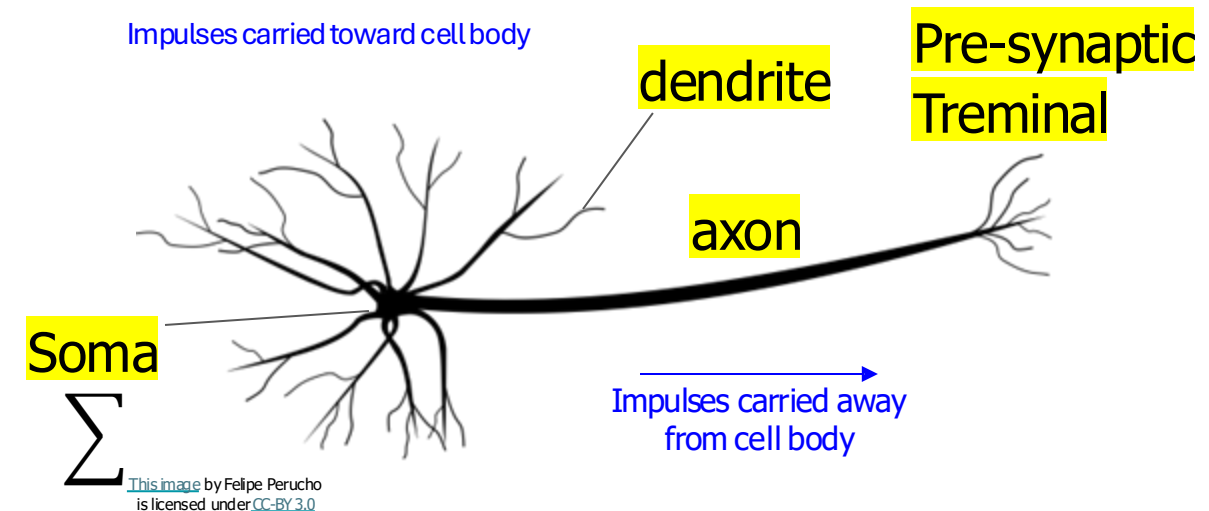
And many other applications, e.g., chess with Deep Reinforcement Learning.



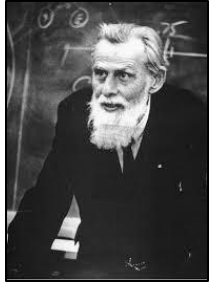
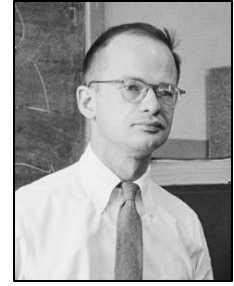
Background: where it started?

The Brain Motivation

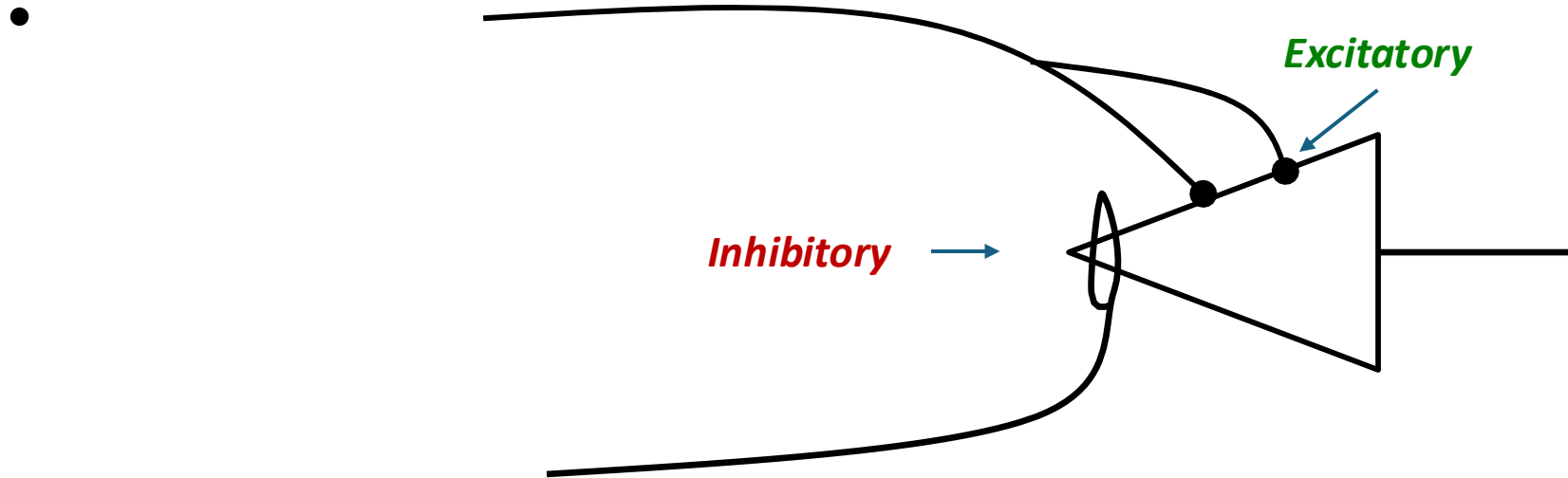
- Mid 1800s: The brain is a mass of interconnected neurons
- A neuron:
 - Signals enter the neuron via the dendrites into the soma
 - Exit through the axon to other neurons



The McCulloch and Pitts model (1943)



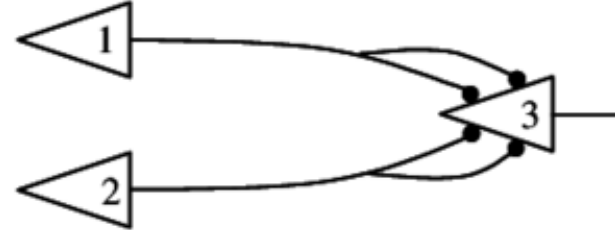
- A mathematical model of a neuron, neuron must receive two excitatory inputs and no inhibitory inputs



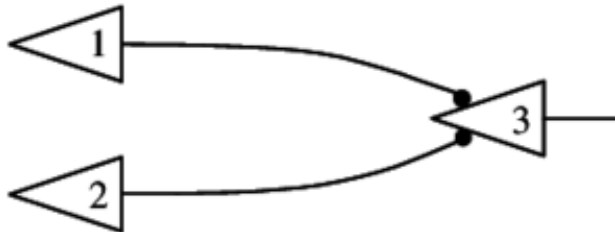
McCulloch and Pitts nets



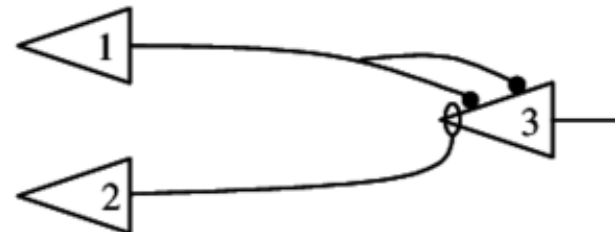
$N_2(t) \Leftrightarrow N_1(t-1)$
net for temporal predecessor



$N_3(t) \Leftrightarrow N_1(t-1) \vee N_2(t-1)$
net for disjunction



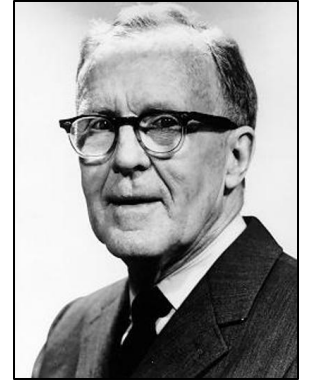
$N_3(t) \Leftrightarrow N_1(t-1) \& N_2(t-1)$
net for conjunction



$N_3(t) \Leftrightarrow N_1(t-1) \& \sim N_2(t-1)$
net for conjunction and negation

However, they didn't provide a learning mechanism....

Donald Hebb model (1949)



Learning via correlations:

“When an axon of cell A is near enough to excite a cell B and repeatedly or persistently takes part in firing it, some growth process or metabolic change takes place in one or both cells such that A's efficiency, as one of the cells firing B, is increased.”

(Neurons that fire together wire together)

- A mathematical model:
 - $x \in \mathbb{R}^d$ a feature input vector
 - $w \in \mathbb{R}^d$ weights for one neuron
 - $y \in \mathbb{R}$ scalar output of the neuron
- $$y = w^\top x, \quad \Delta w = \eta(xy)$$
- When x (presynaptic signal) is large **and** y (postsynaptic firing) is large, then Δw is large (η is the learning rate).

Hebbian Learning

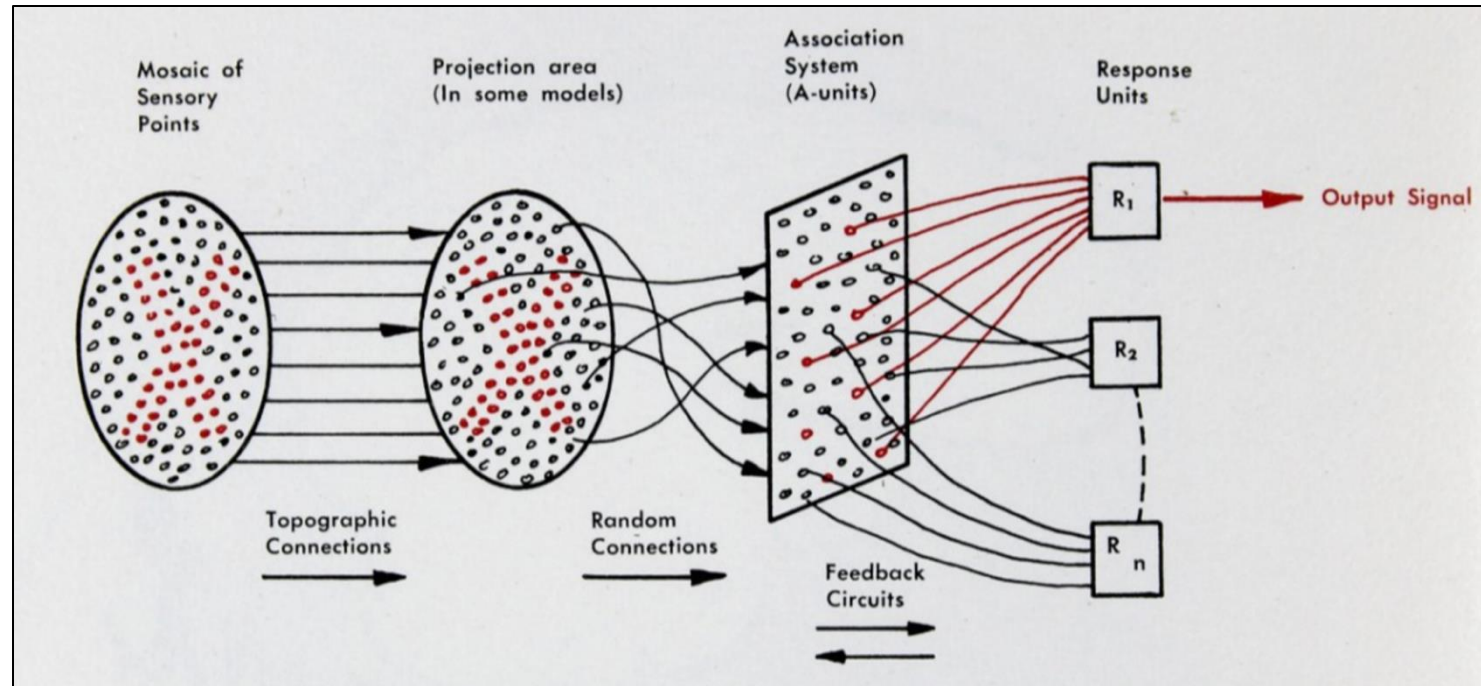
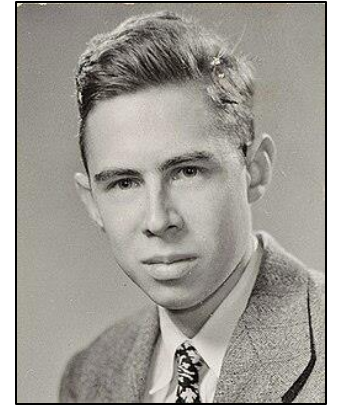
- Stability issues:
 - No reduction in weights
 - Learning is unbounded
- Follow up works allowing for weight normalization, forgetting etc.
- E.g., Oja's rule:

$$y = w^T x, \quad \Delta w = \eta y(x - yw)$$

- Increase in proportion to the correlation with x
- Decrease when the neuron's output is large

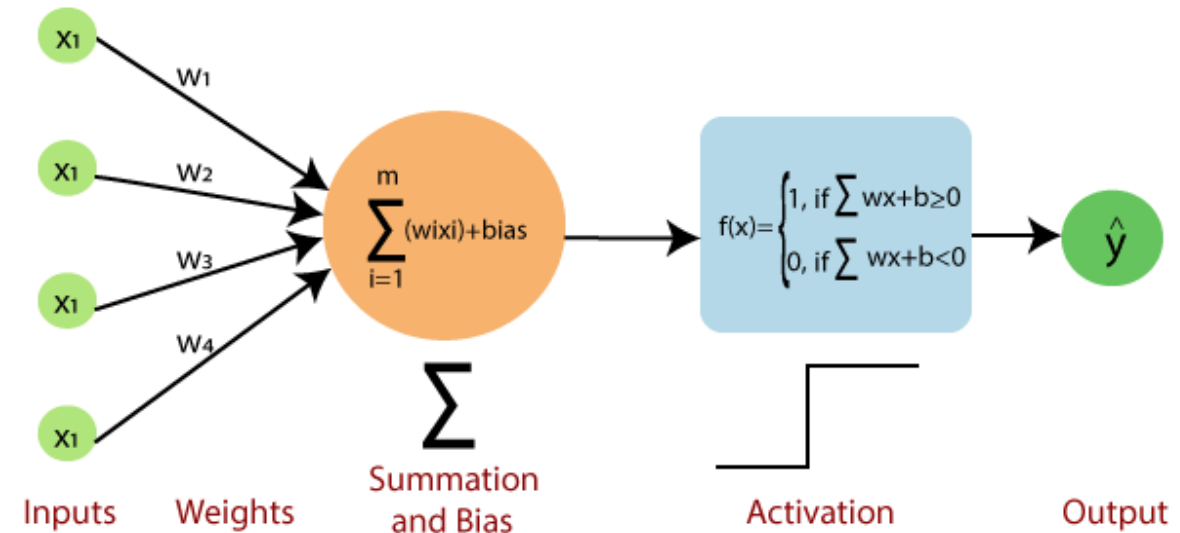
Rosenblatt's Perceptron (1958)

- **Learns from labels!**



The Perceptron

- Input: feature vector $x \in \mathbb{R}^m$
- Label: $y \in \{0, 1\}$
- Model:
$$\hat{y} = \sigma(w^T x + b)$$
- **Update rule:** $\Delta w = \eta(y - \hat{y})x$
- The learning rate η



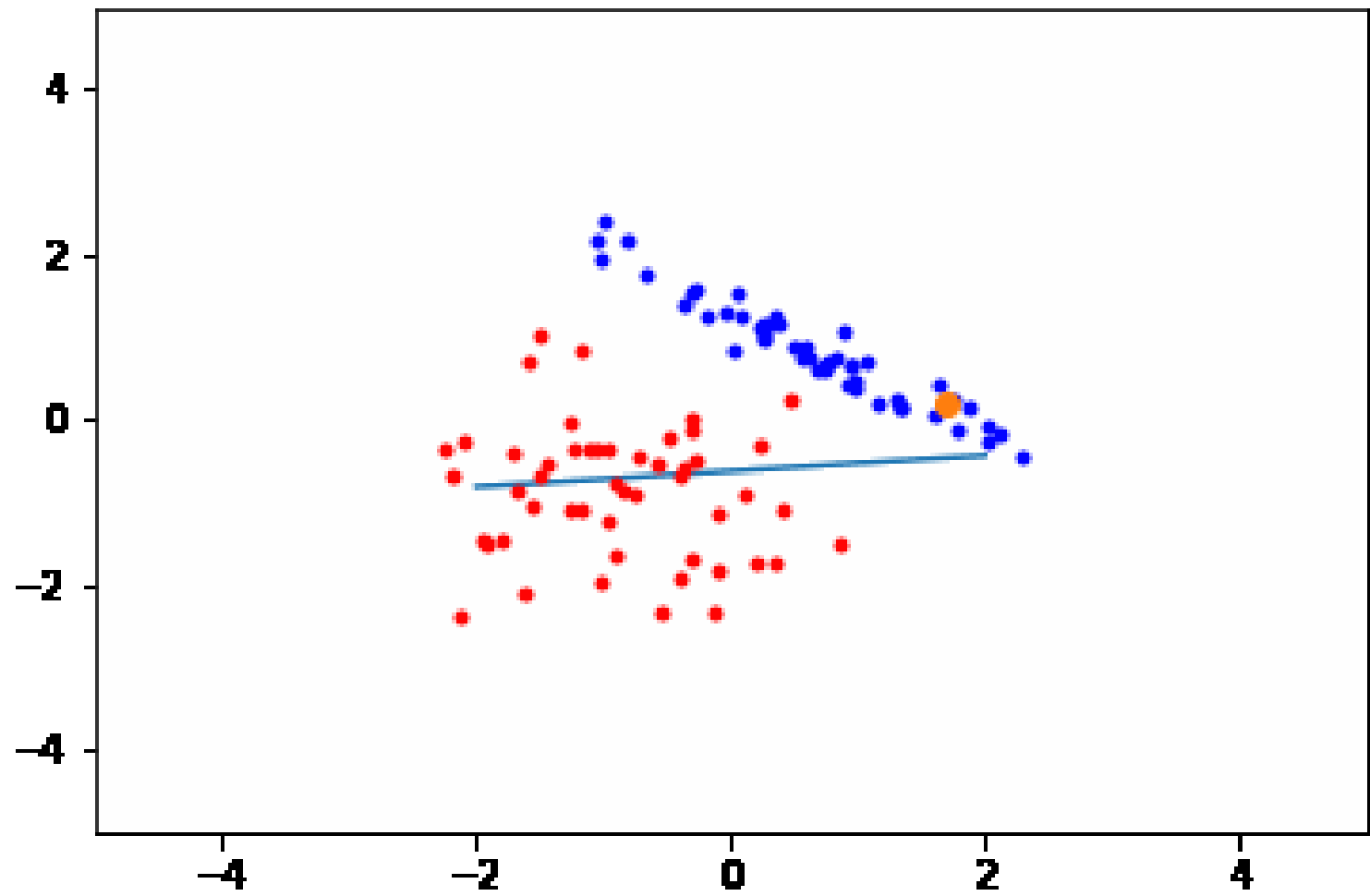
Learning intuition

$$\Delta w = \eta(y - \hat{y})x$$

Geometrically: we want a **hyperplane** (defined by w) that separates positive from negative examples.

How it learns?

- If $\hat{y} = y$, then no update
- If $\hat{y} = 0$ and $y = 1$ (false negative), then w move to include x
$$w \leftarrow w + (1 - 0)x$$
- If $\hat{y} = 1$ and $y = 0$ (false positive), then w move to exclude x
$$w \leftarrow w + (0 - 1)x$$

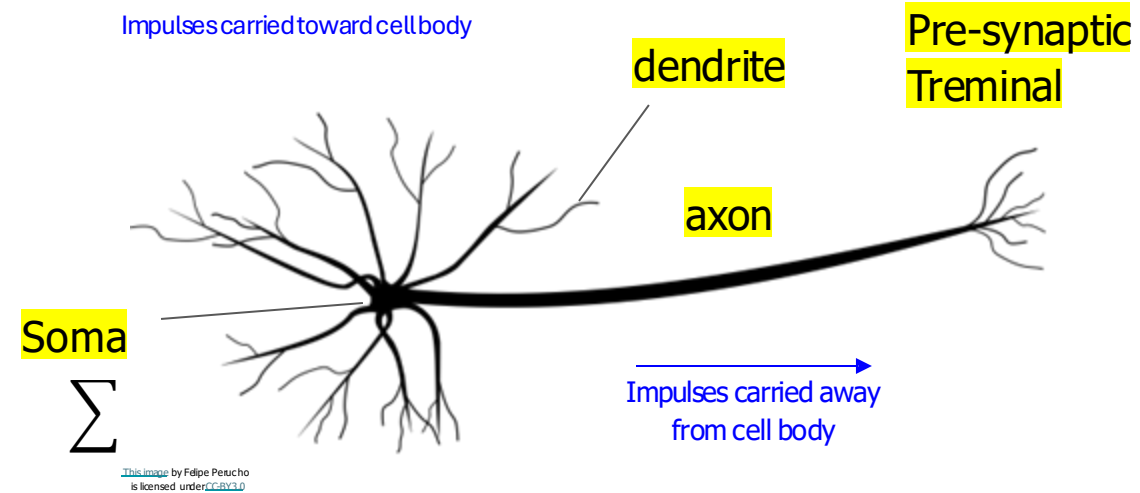


Human-like capabilities



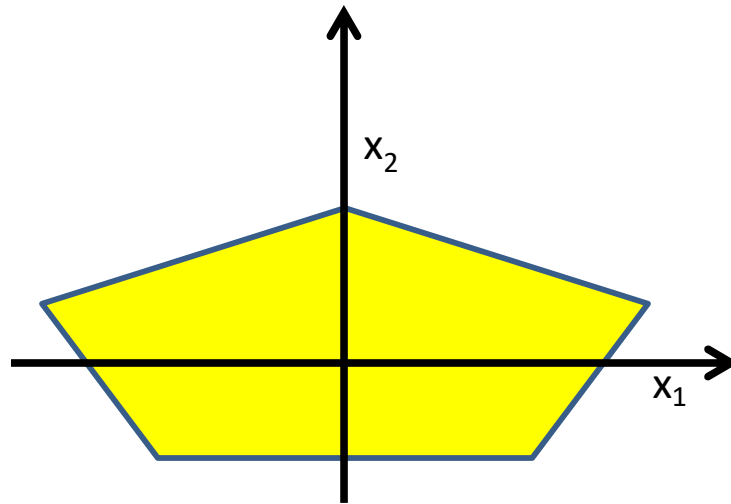
Real Neurons vs Perceptron

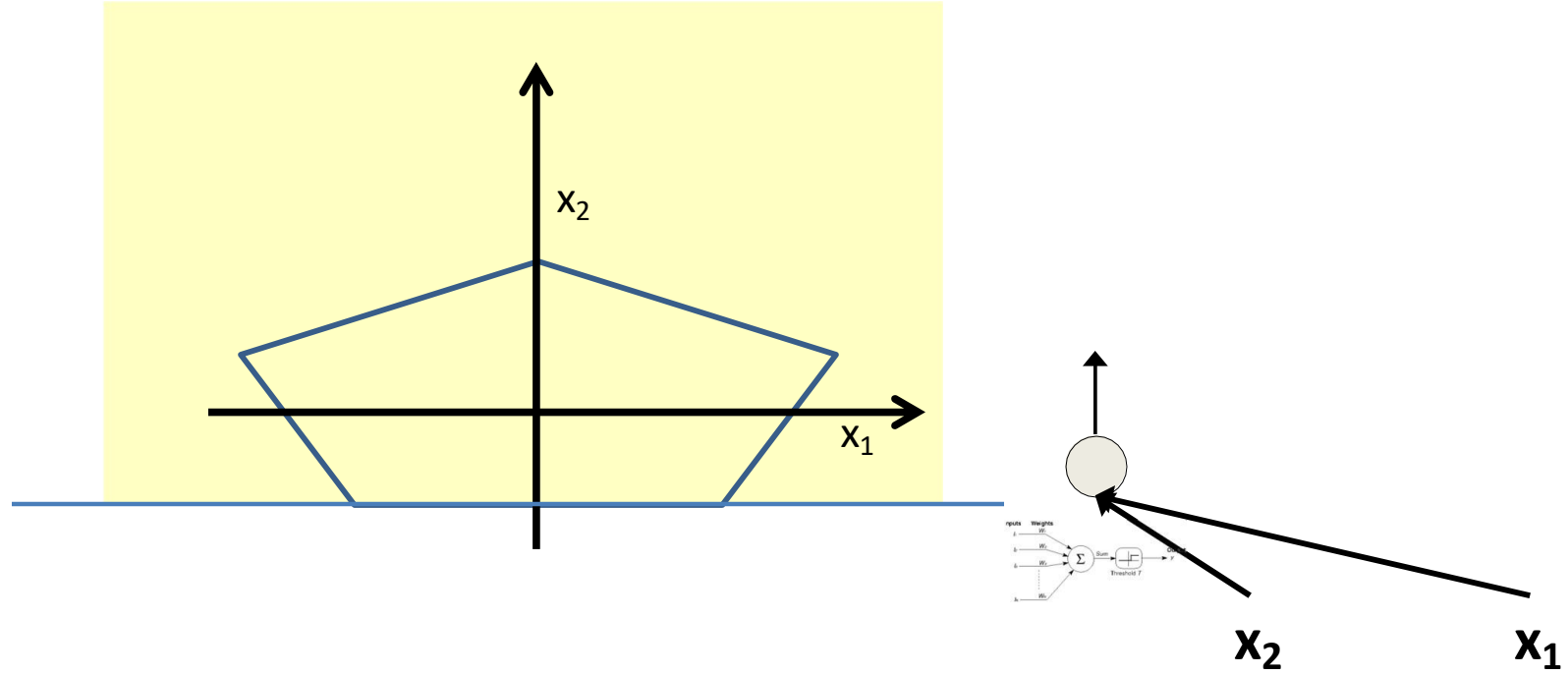
- Many neuron types
 - sensory, inhibitory, excitatory; each with unique roles.
- Dendrites compute non-linearly
 - amplify some signals, cancel others.
- Synapses are dynamic
 - weights change with timing, frequency, and chemicals.

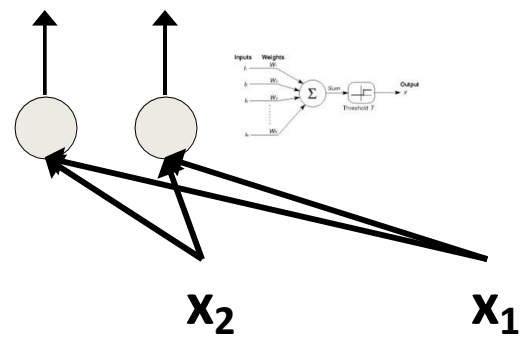
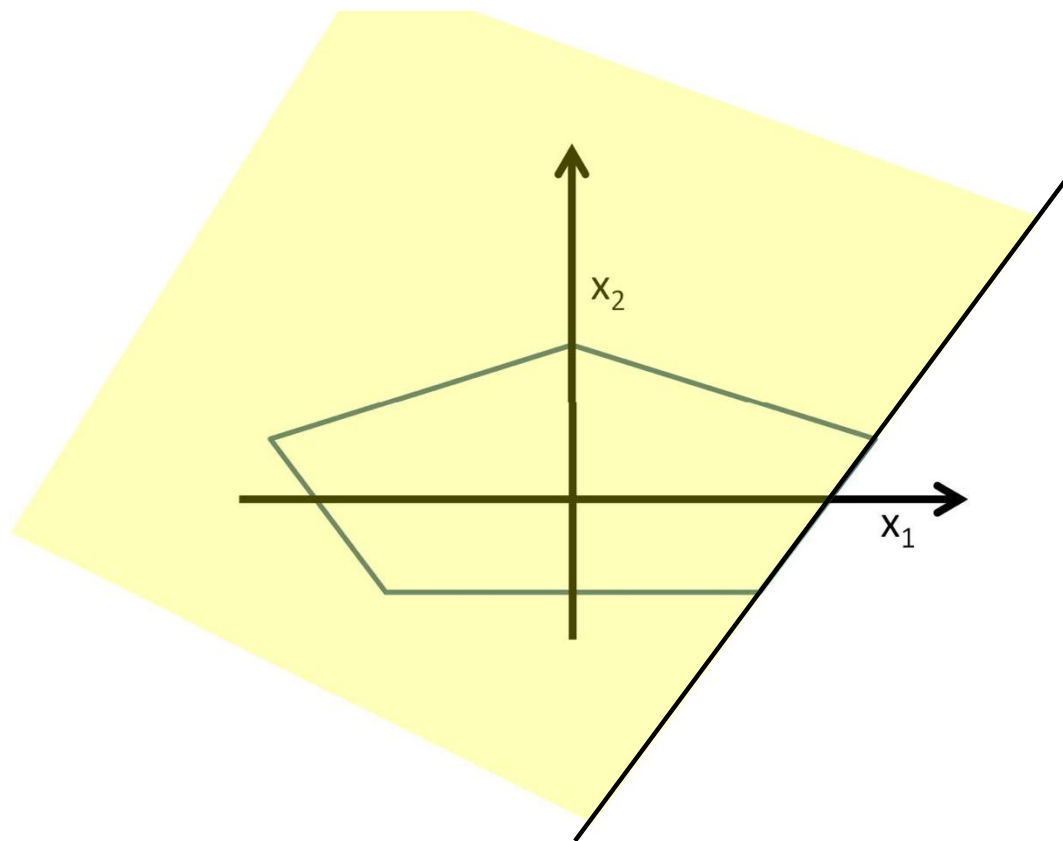


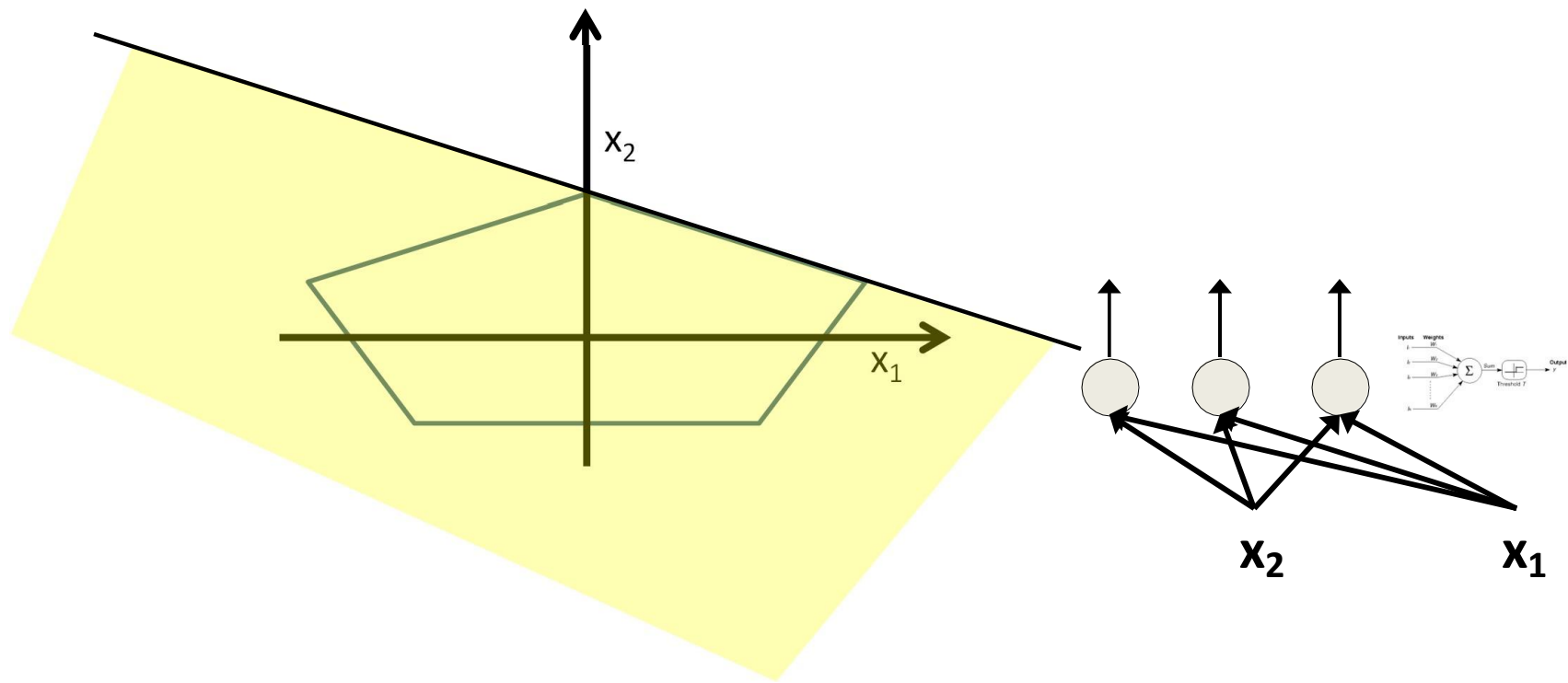
Composing complicated “decision” boundaries

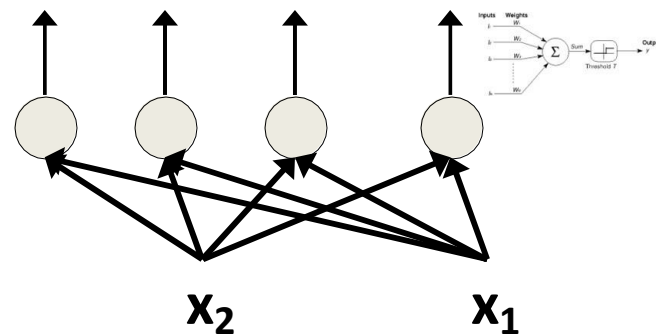
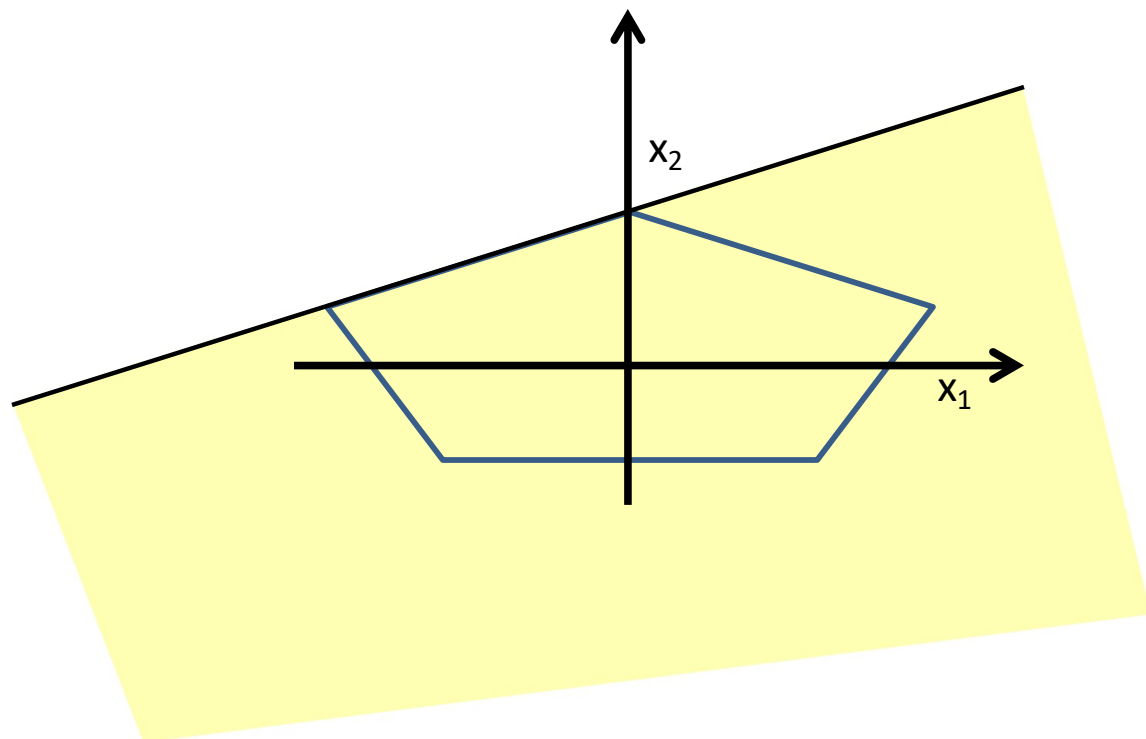
- Build a network of units with a single output that fires if the input is in the colored area

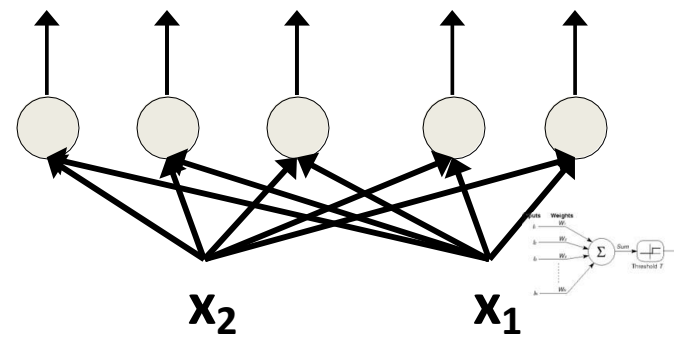
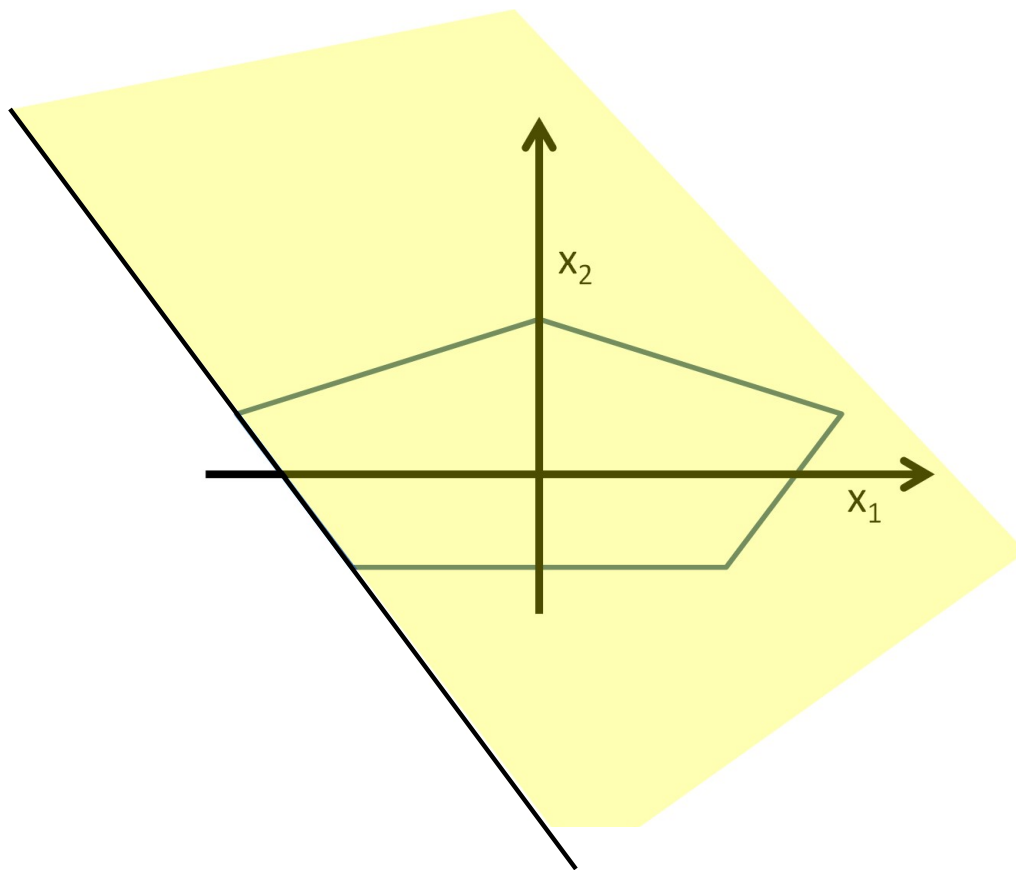


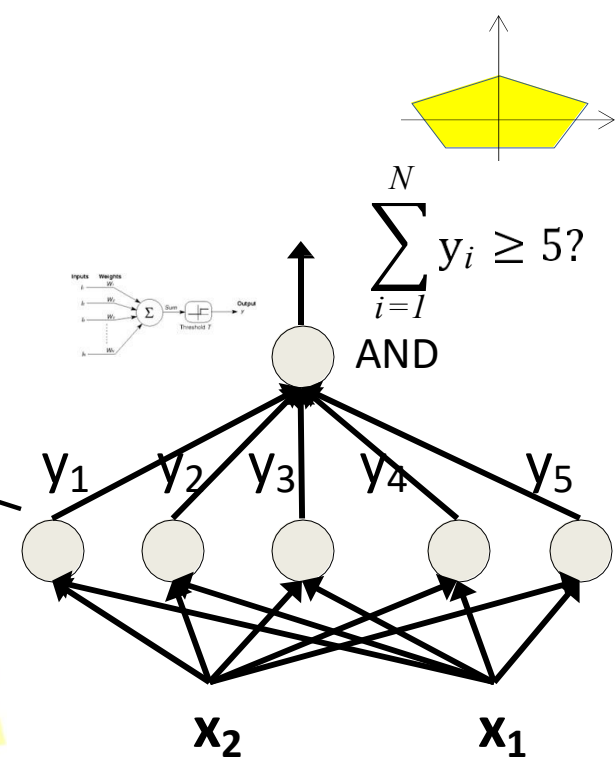
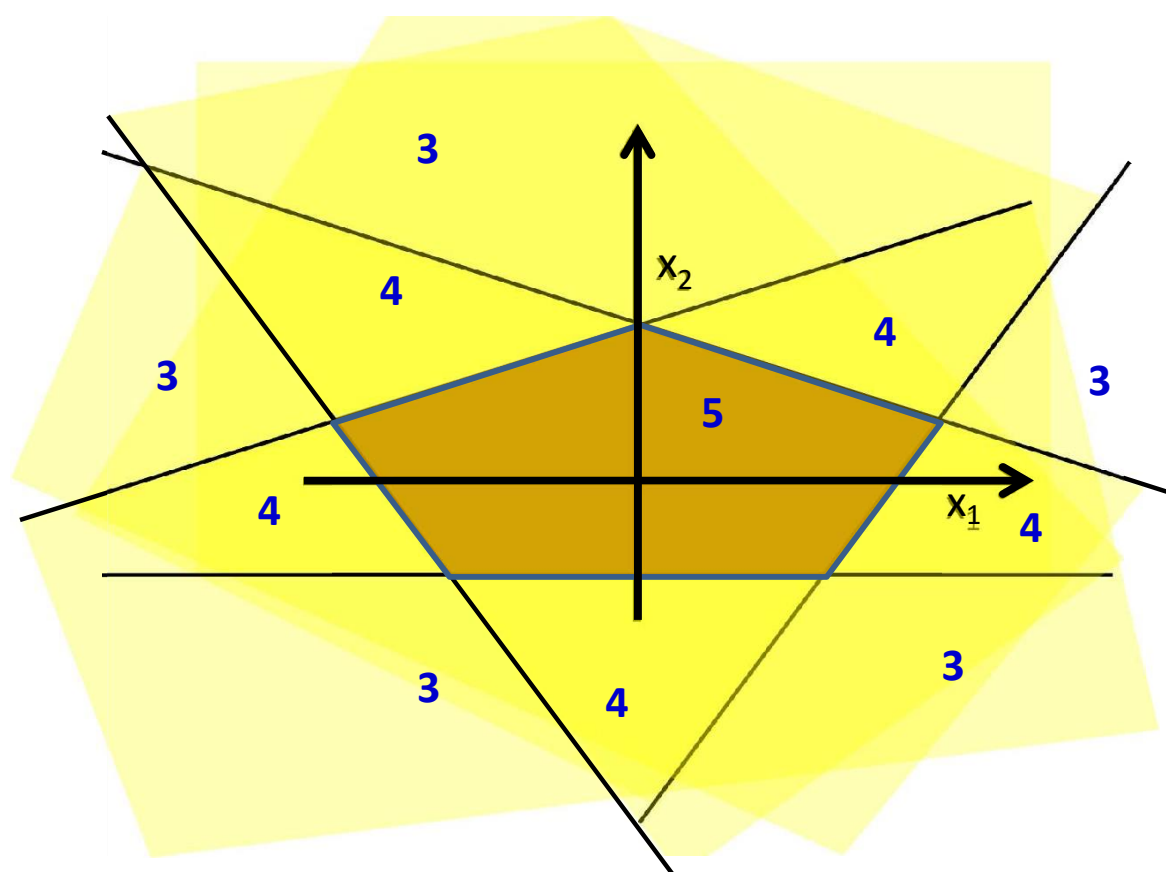






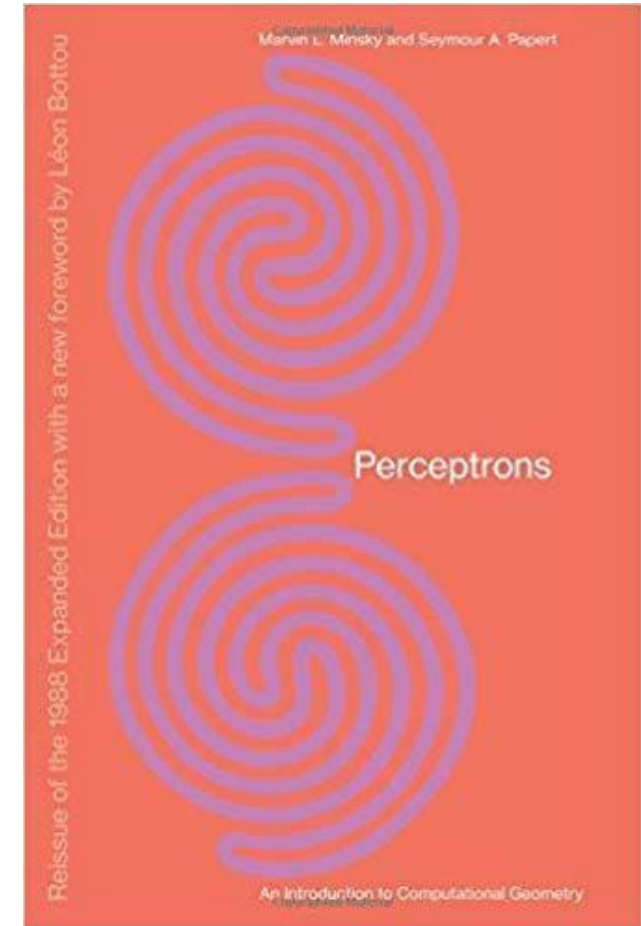
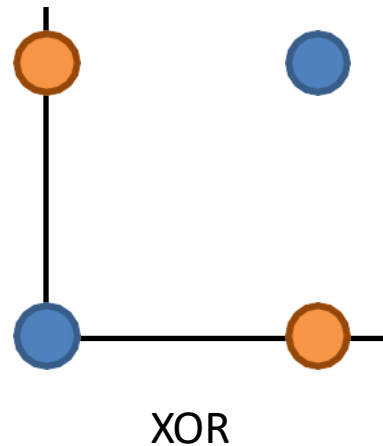






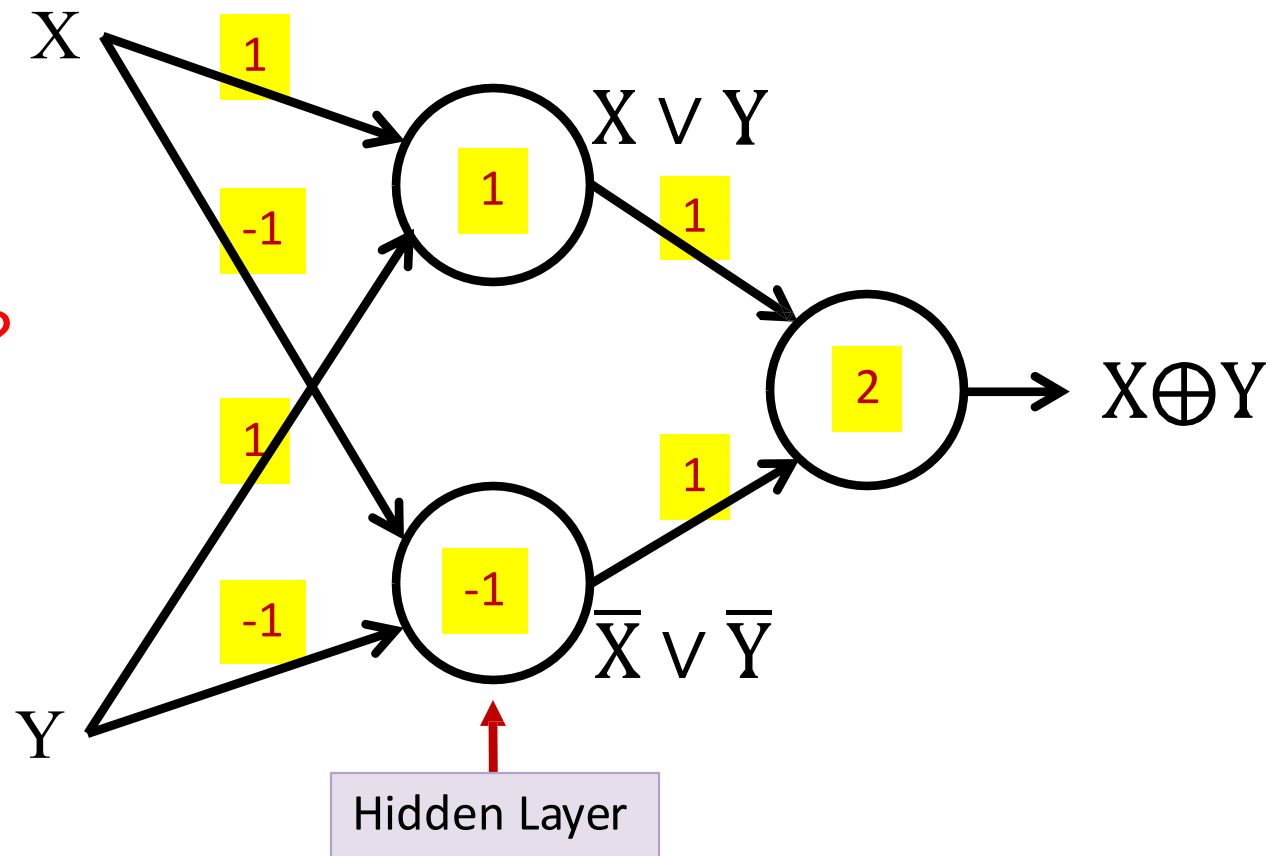
Minsky and Papert, 1969 (AI Winter)

- Showed that Perceptron could not learn the XOR function

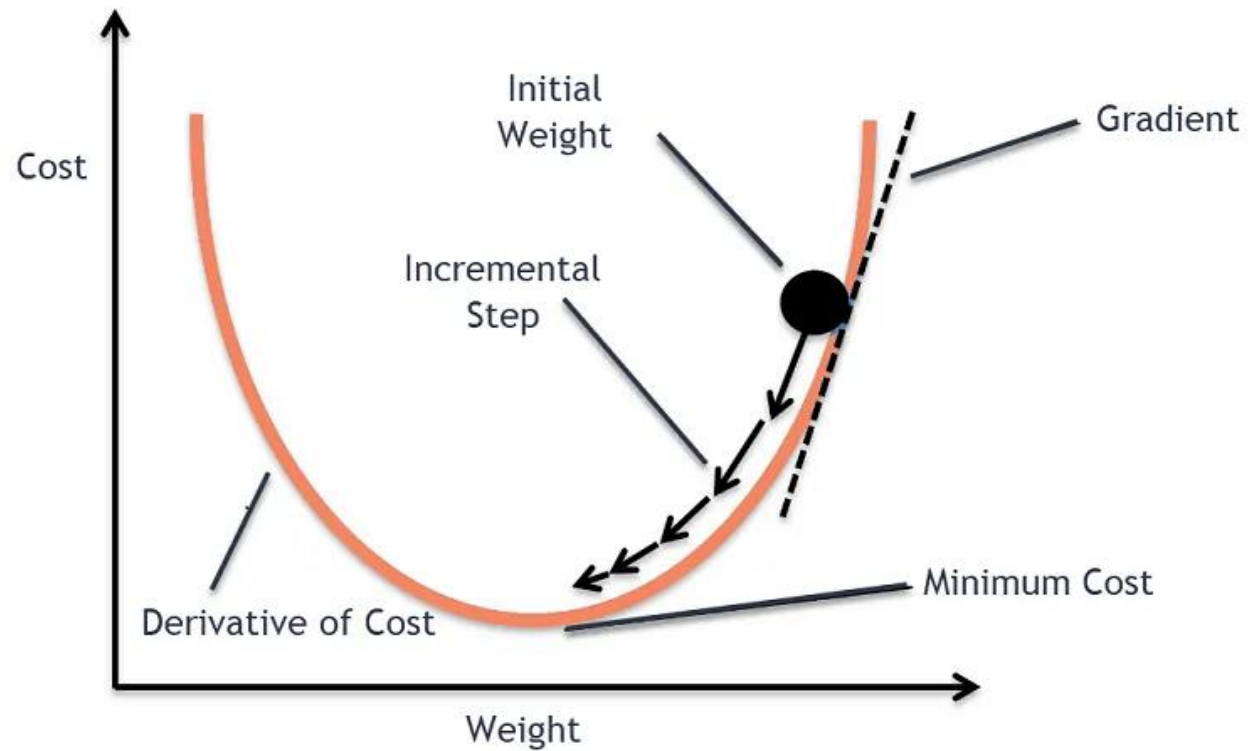
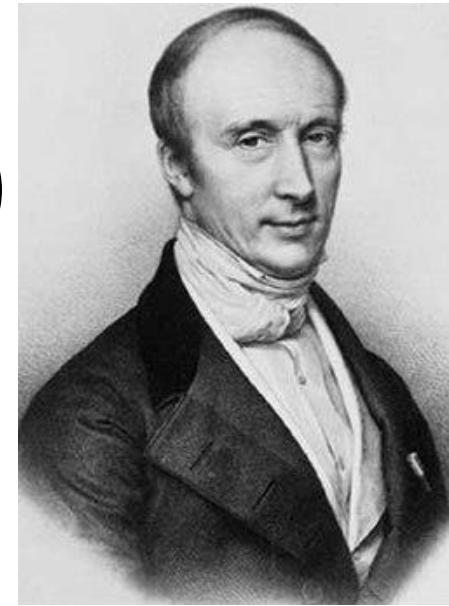


XOR and Multi-layer Perceptron (MLP)

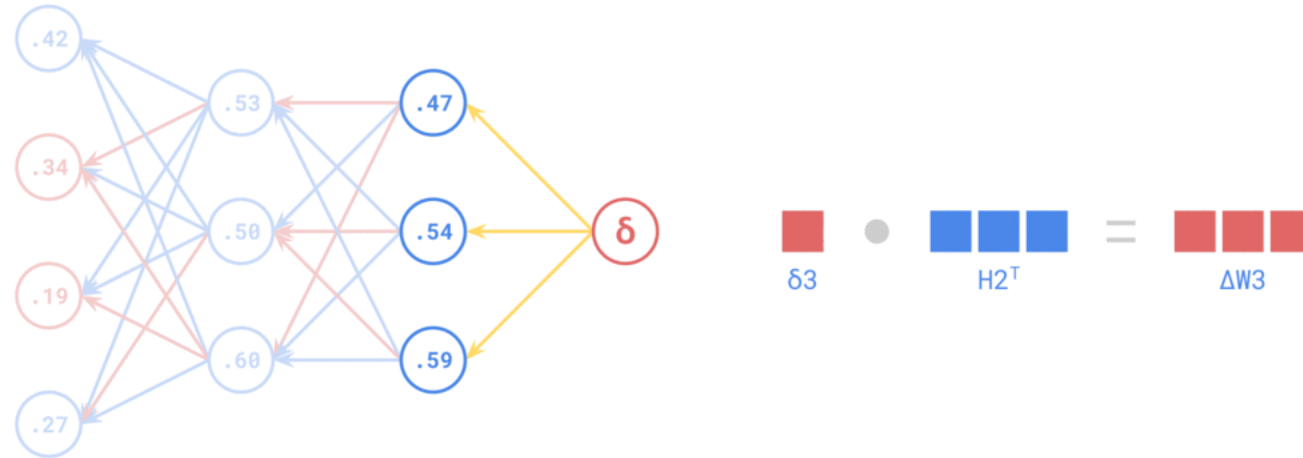
But how to train?



Cauchy and the Gradient Method (1857)



Backpropagation for MLP



Development of Backpropagation

- Optimization methods based on chain rule
 - Gradient Theory of Optimal Flight Paths [**Kelley 1960**]
- Generic and efficient derivations
 - Taylor expansion of the accumulated rounding error [**Linnainmaa 1976**]
- First connection the neural networks
 - **P. J. Werbos** Thesis **1974**
 - System Modeling and Optimization **1981**
- Popularize representation learning
 - Learning representations by back-propagating errors [DE. Rumelhart, **GE. Hinton**, RJ. Williams **1986**]



Paul Werbos

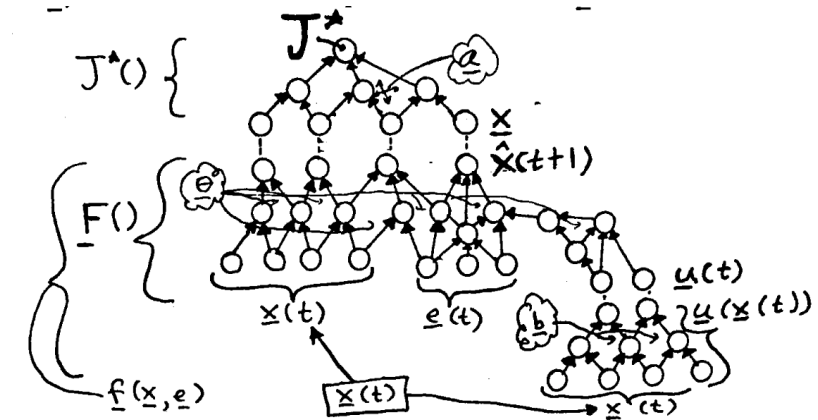


Figure 5: Realization of GDHP As a Triple Network to Make Decisions

Figure from System
Modeling and Optimization

Is Backpropagation the solution?

Forward-Forward; Hinton'23

- Instead of backpropagating errors (like in backpropagation), FF trains by comparing activations under two conditions:
 - Positive data: real examples from the dataset
 - Negative data: corrupted data
- The **goodness function** quantifies how “good” the hidden representations

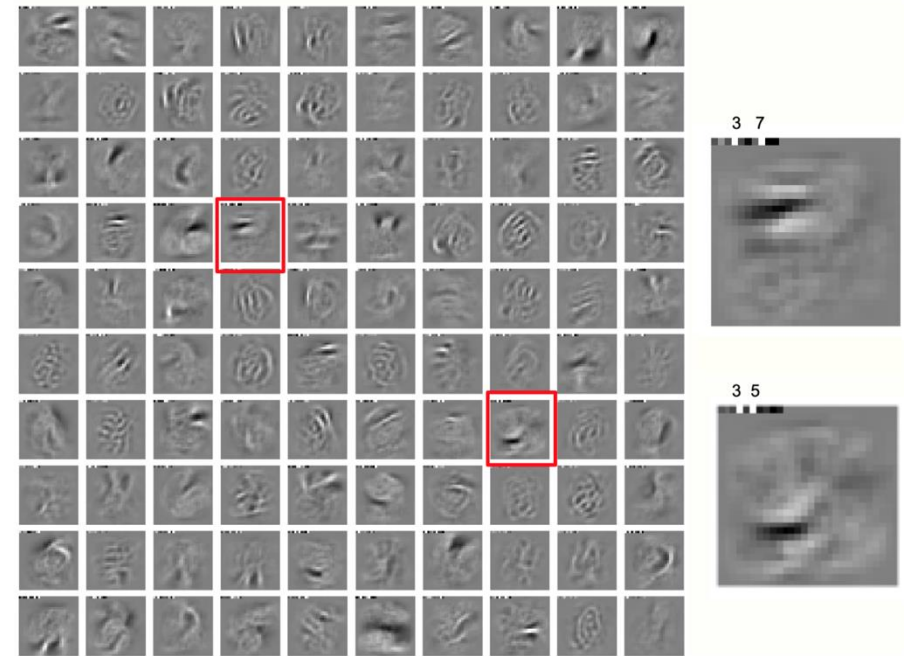
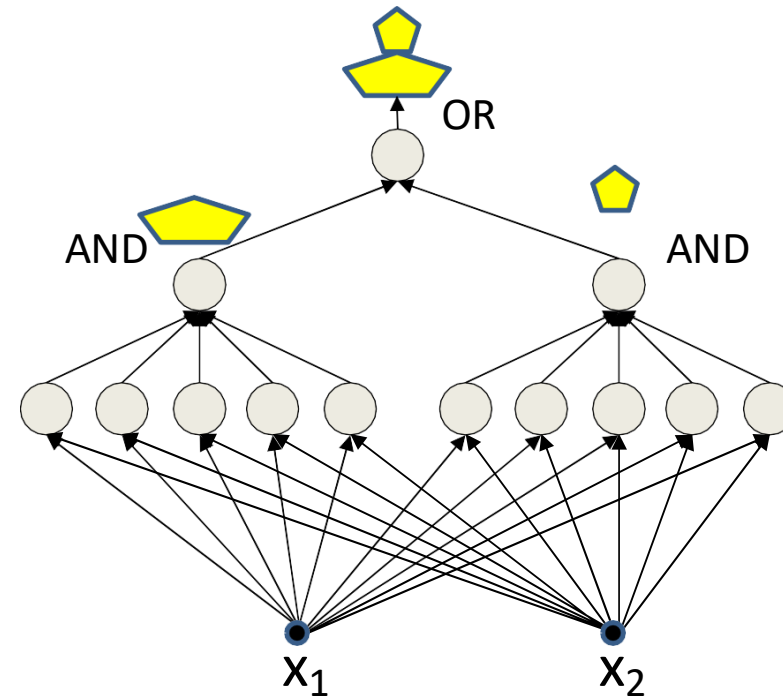
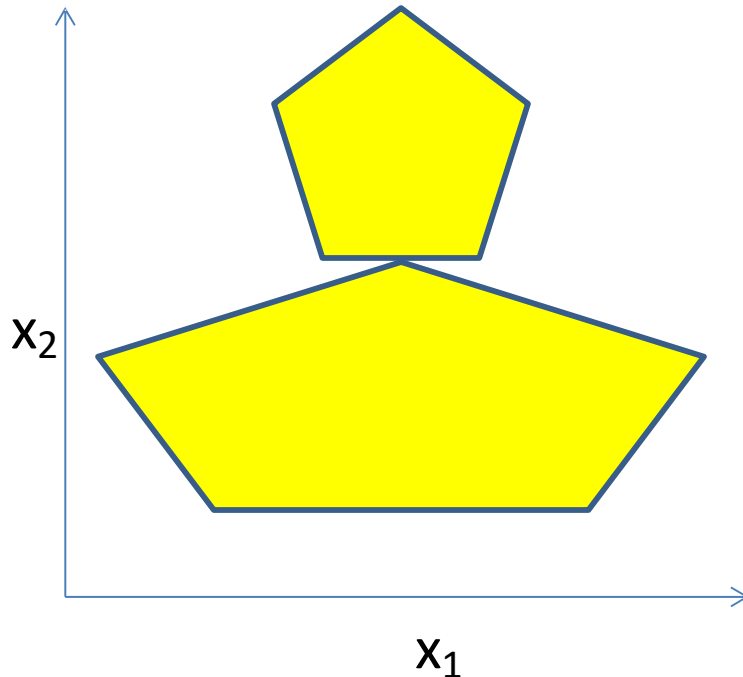


Figure 2: The receptive fields of 100 neurons in the first hidden layer of the network trained on jittered MNIST. The class labels are represented in the first 10 pixels of each image.

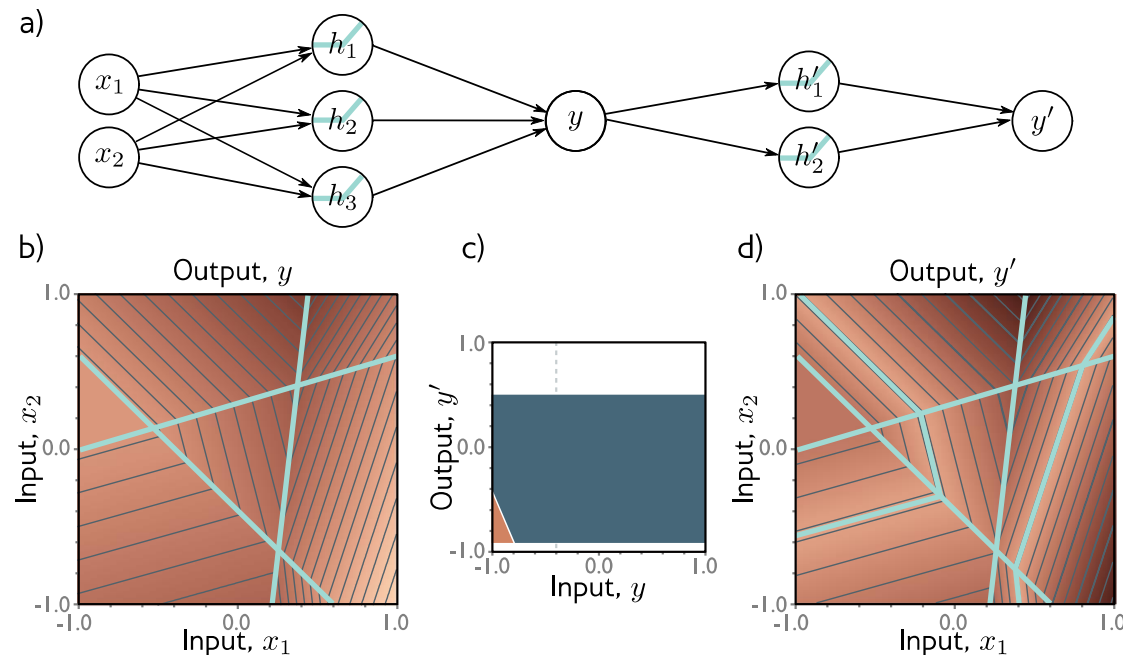
MLP – More complex decision boundaries

- Network to fire if the input is in the yellow area
 - “OR” two polygons
 - A third layer is required

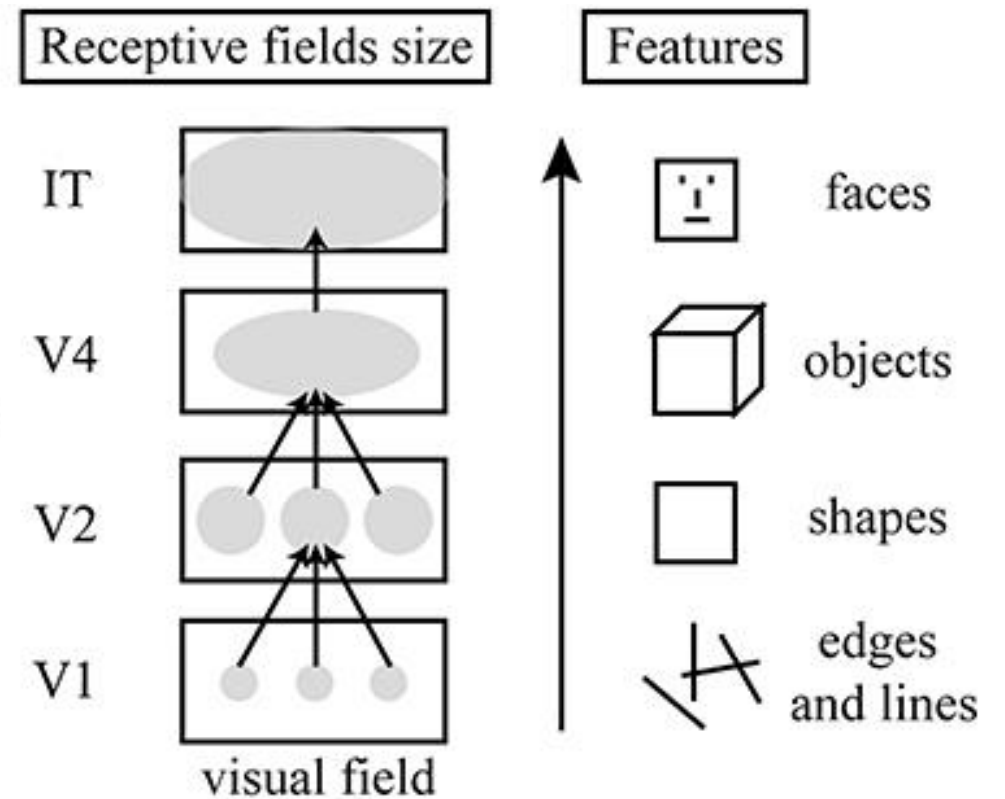
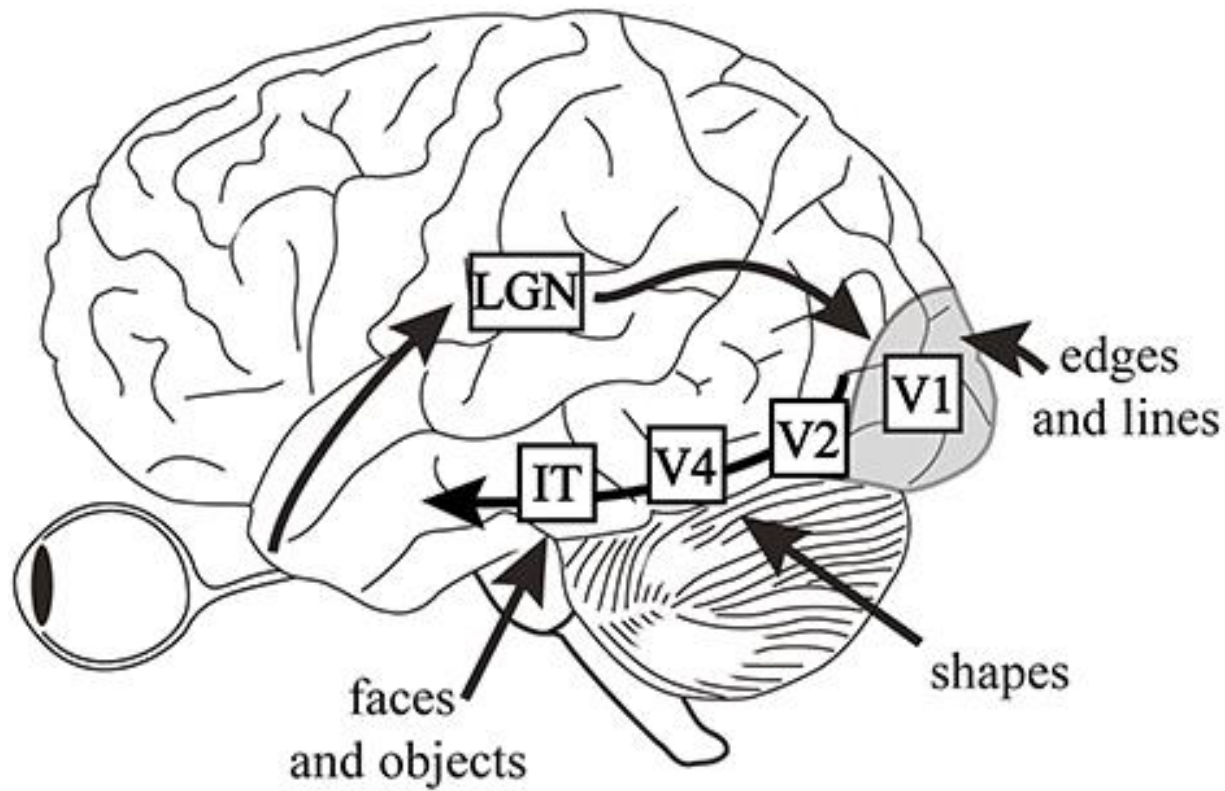


Depth efficiency

- Deep networks create many more regions per parameters
- But there are dependencies between them



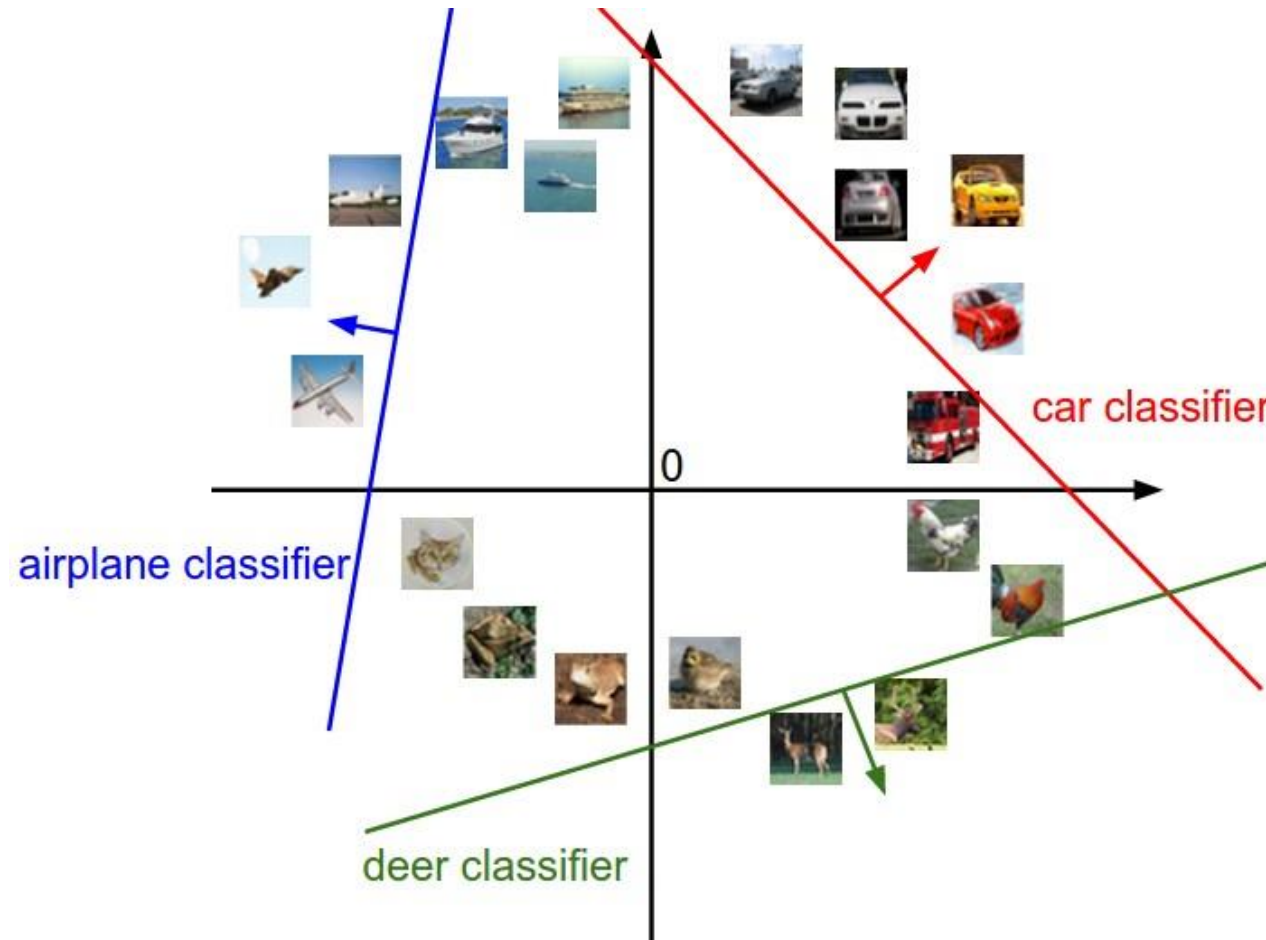
Deep learning and connection to the Visual Cortex



Hubel & Wiesel Experiment (1959)

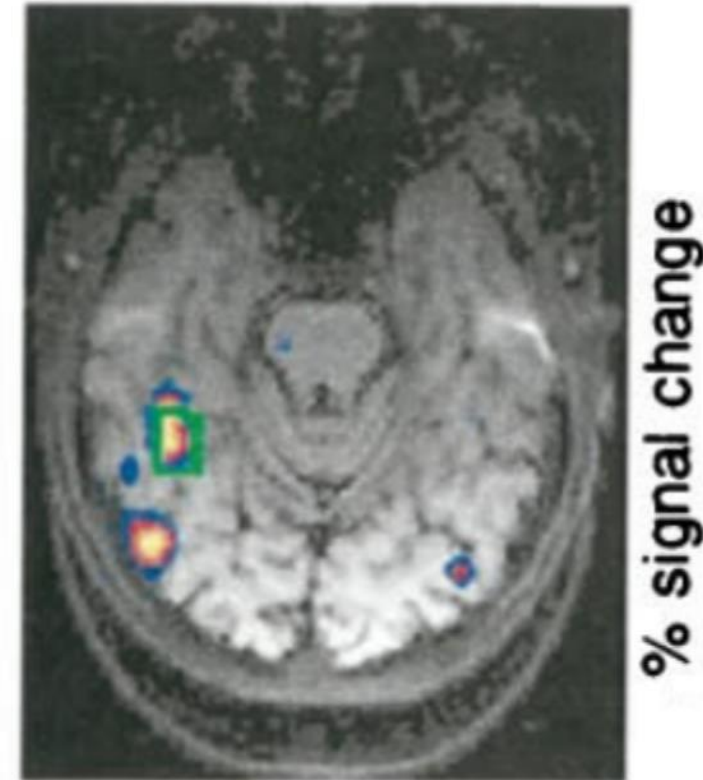
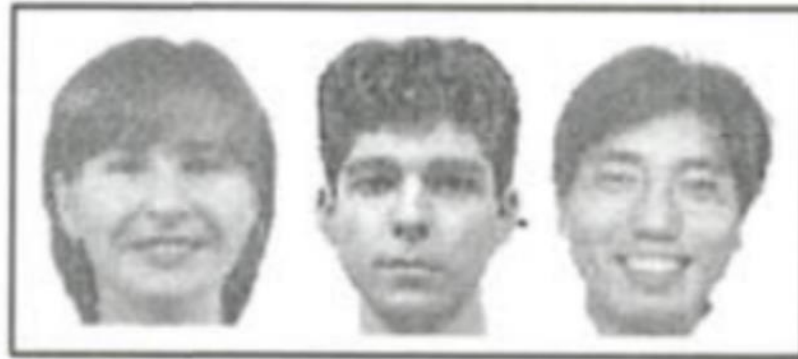


Image Classification?



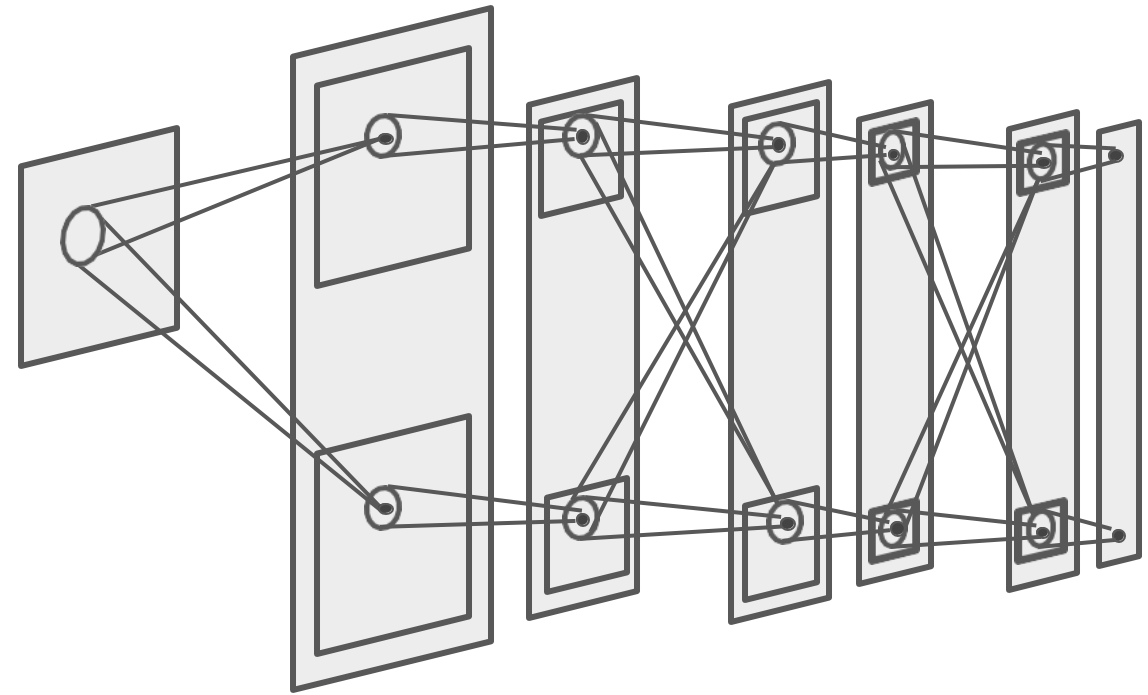
Neural correlation to abstract concepts (Kanwisher et al., 1997)

Faces > Houses

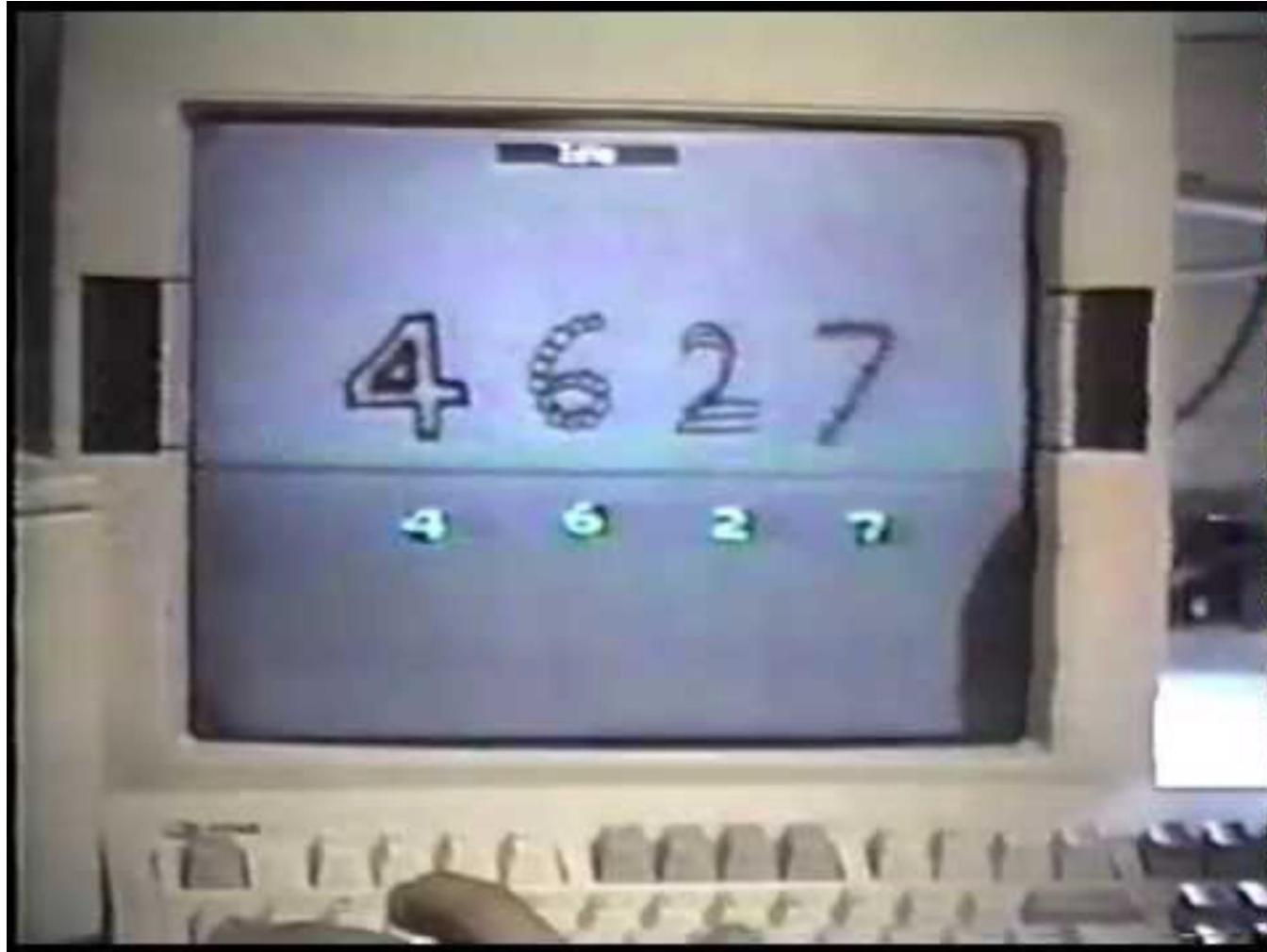


Neocognitron: Fukushima, 1980

- Computational model the visual system, directly inspired by Hubel and Wiesel's hierarchy of complex and simple cells
- Interleaved simple cells (convolution) and complex cells (pooling)
- No practical training algorithm

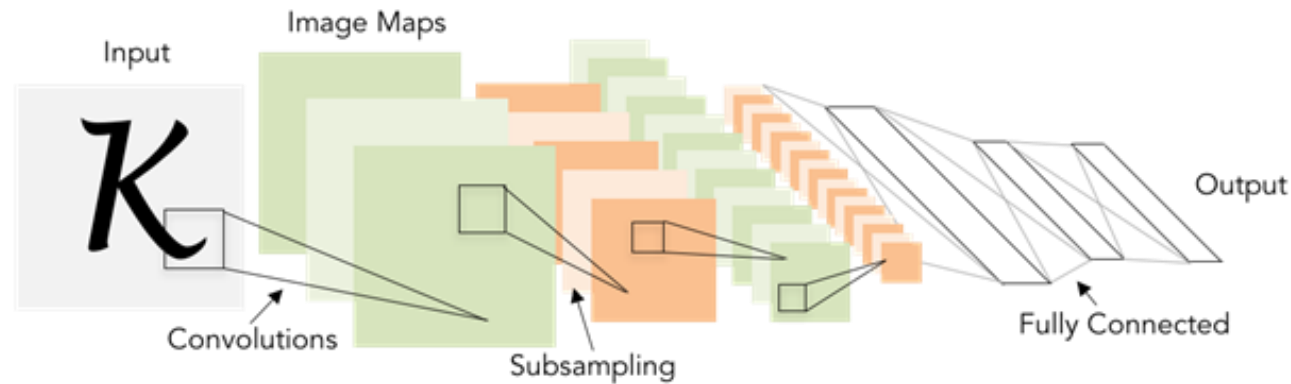


Backpropagation Applied to Handwritten Zip Code Recognition [LeNet-1; NIPS'89]



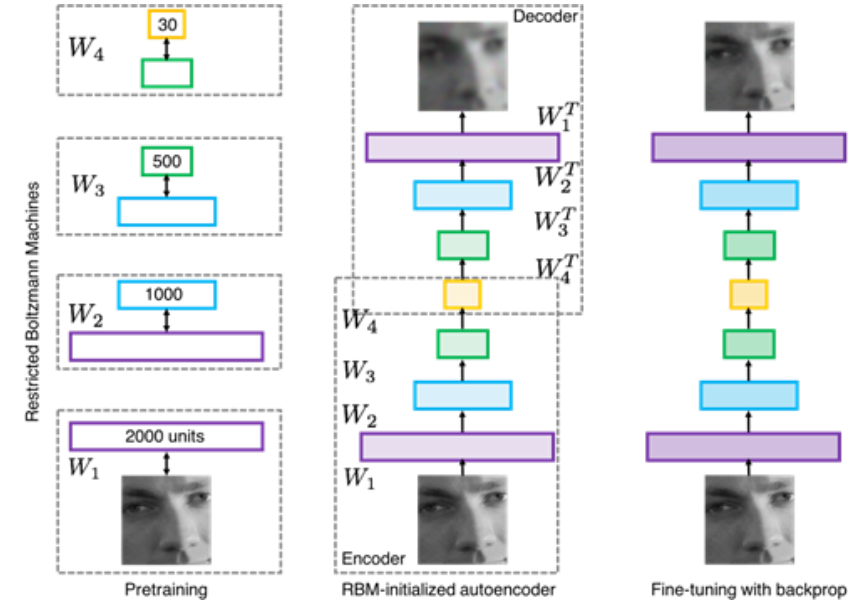
LeNet-5 (LeCun et al., 1998)

- Applied backprop algorithm to a Neocognitron-like architecture
- Learned to recognize handwritten digits
- Was deployed in a commercial system by NEC
- Very similar to our modern convolutional networks!



2000s: “Deep Learning”

- Exists but not a mainstream research topic
 - No benchmarks
 - No Datasets

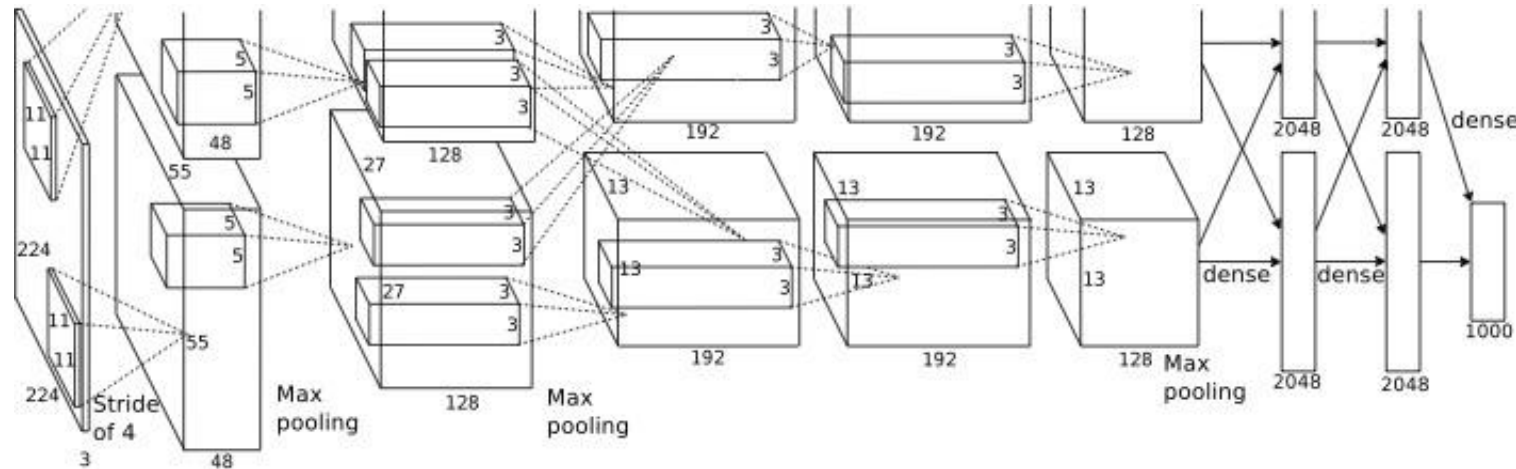


Hinton and Salakhutdinov, 2006

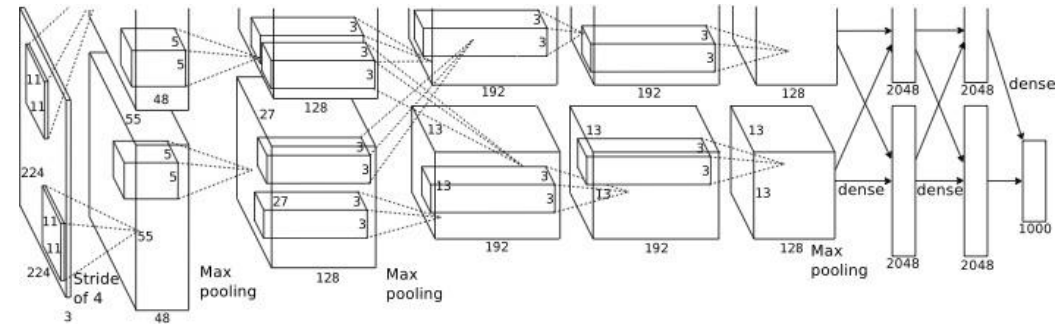
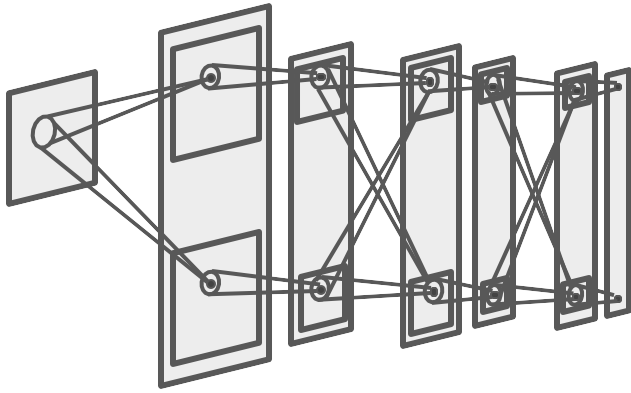
AlexNet (2012)



- Significant improvements on large-scale classification challenge
- Very efficient (ReLU activations, GPU-based optimization)
- Deep learning becomes mainstream



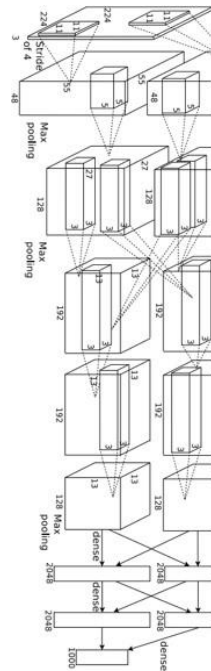
AlexNet vs. Neocognitron: 32 years apart



Continued refinement in image classification

Year 2012

SuperVision

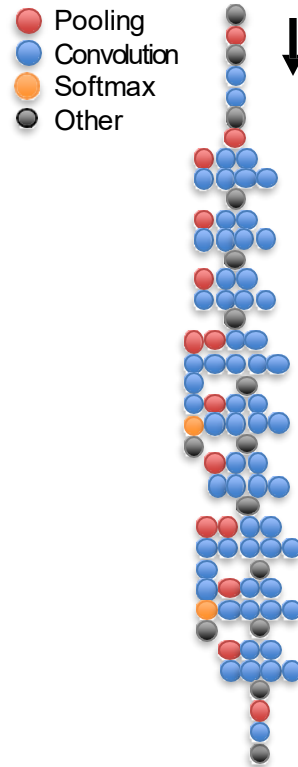


[Krizhevsky NIPS 2012]

Figure copyright Alex Krizhevsky, Ilya Sutskever, and Geoffrey Hinton, 2012. Reproduced with permission.

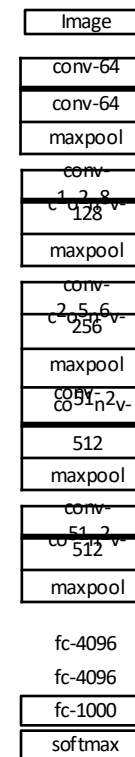
Year 2014

GoogLeNet



[Szegedy arxiv 2014]

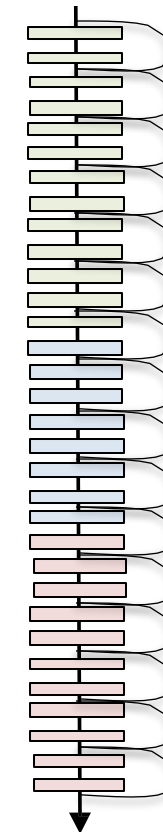
VGG



[Simonyan arxiv 2014]

Year 2015

ResNet



[He ICCV 2015]

Turing Award for deep learning 2018



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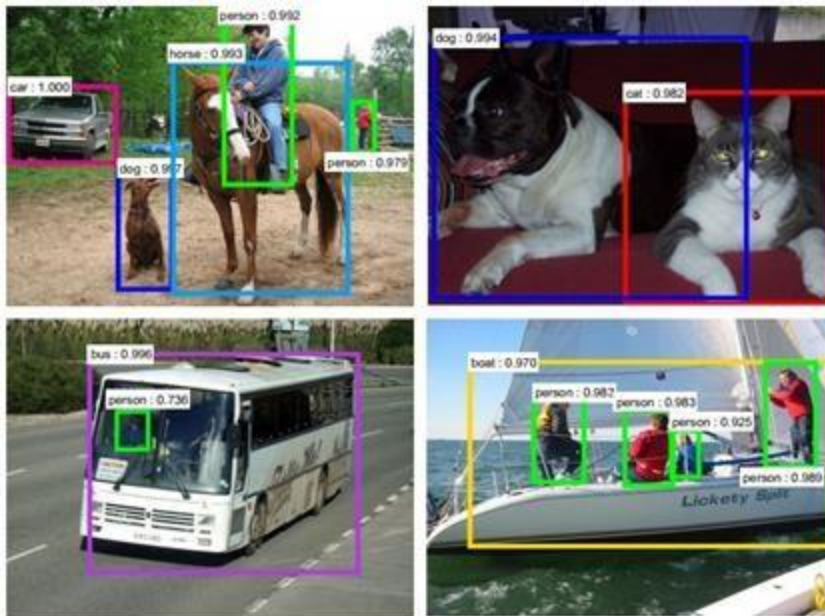
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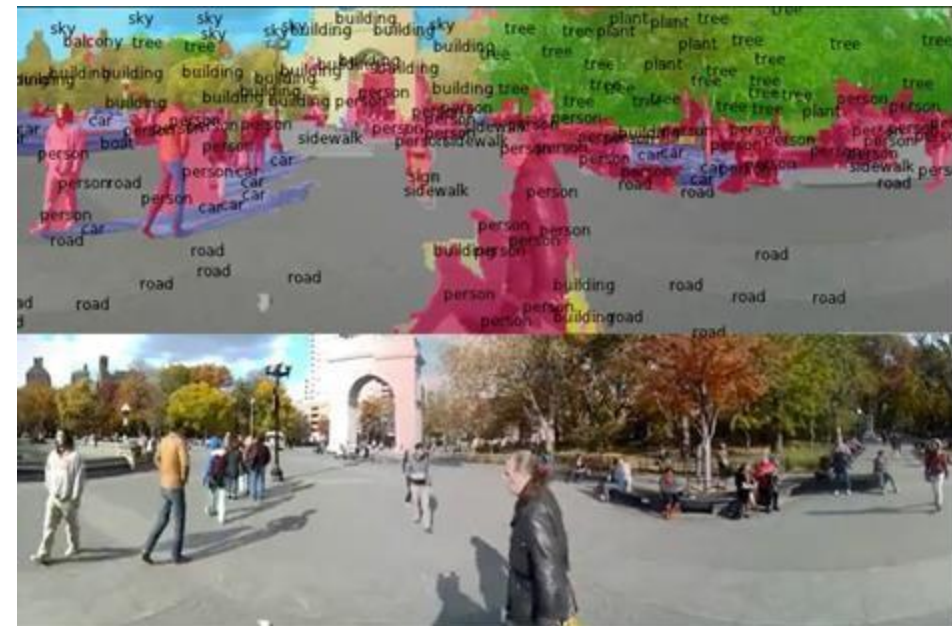
Autonomous Driving 2019

Object Detection



Ren, He, Girshick, and Sun, 2015

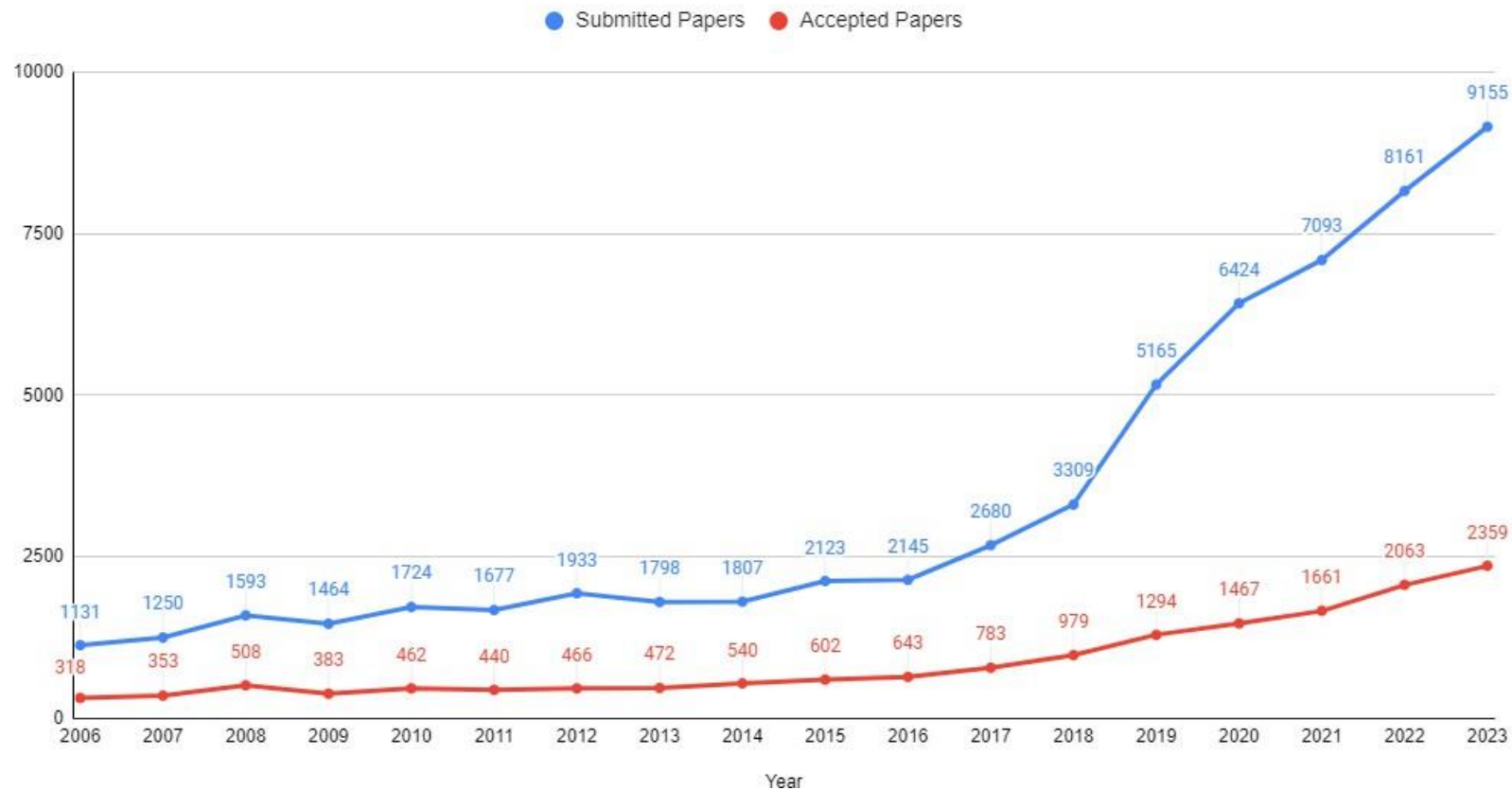
Image Segmentation



Fabaret et al, 2012

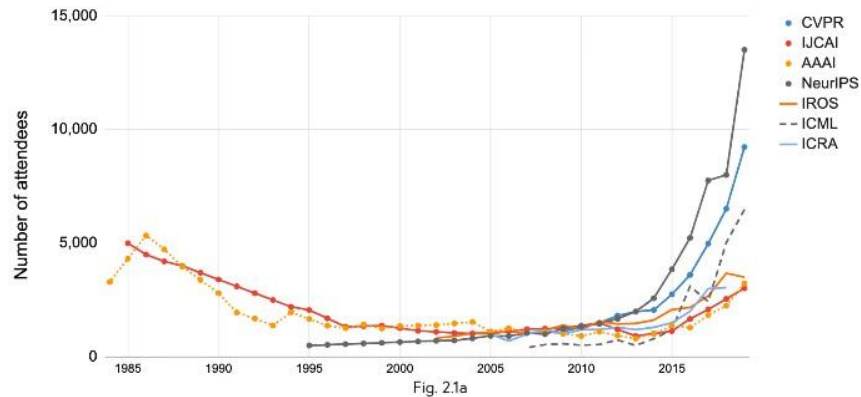
Deep learning explosion in academia

CVPR Conference Paper Statistics



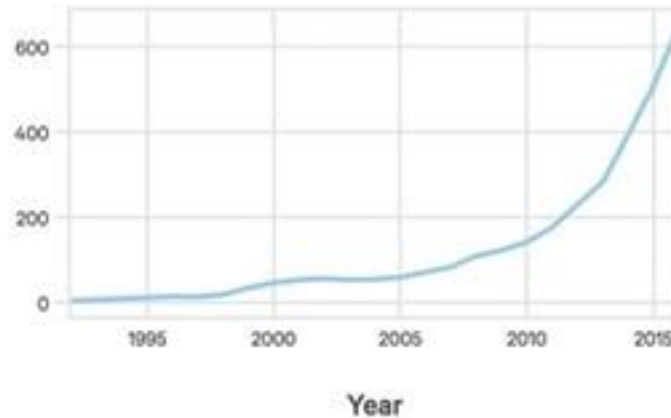
AI's explosive growth & impact

Attendance at large conferences (1984-2019)
Source: Conference provided data.



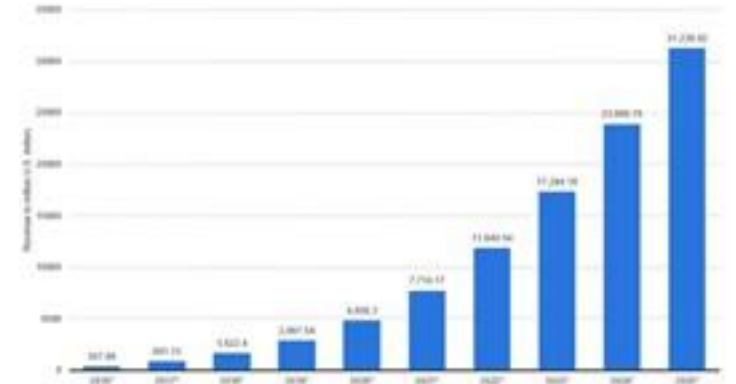
**Number of attendance
At AI conferences**

Source: The Gradient



**Startups Developing AI
Systems**

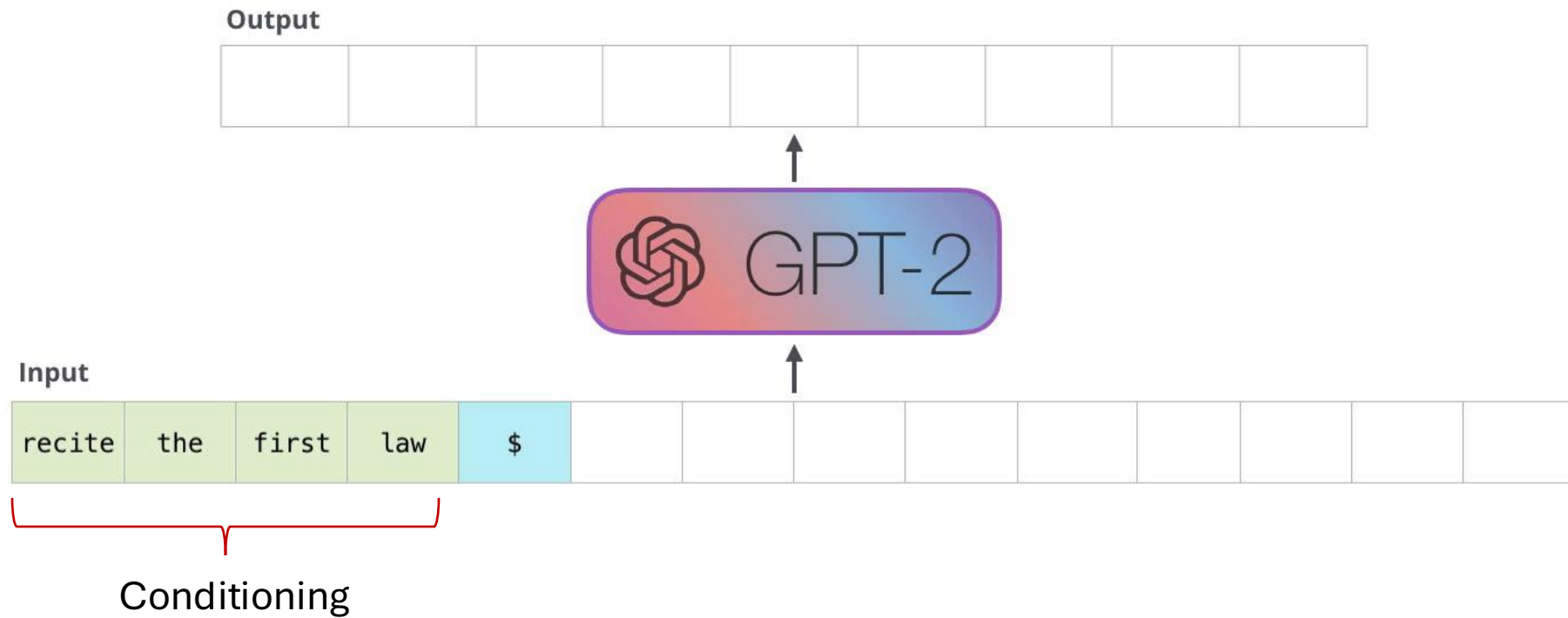
Source: Crunchbase, VentureSource, Sand
Hill Econometrics



**Enterprise Application AI
Revenue**

Source: Statista

Not only for image: Language models 2019



ChatGPT – Learn to follow instructions 2022

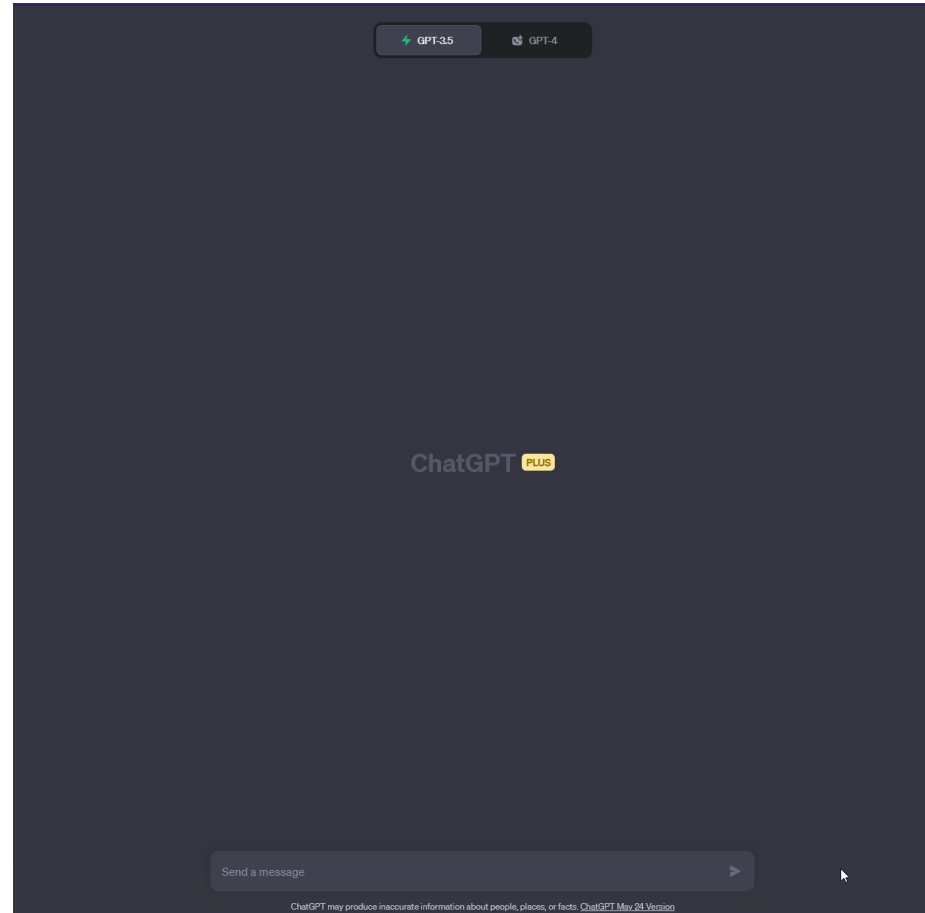
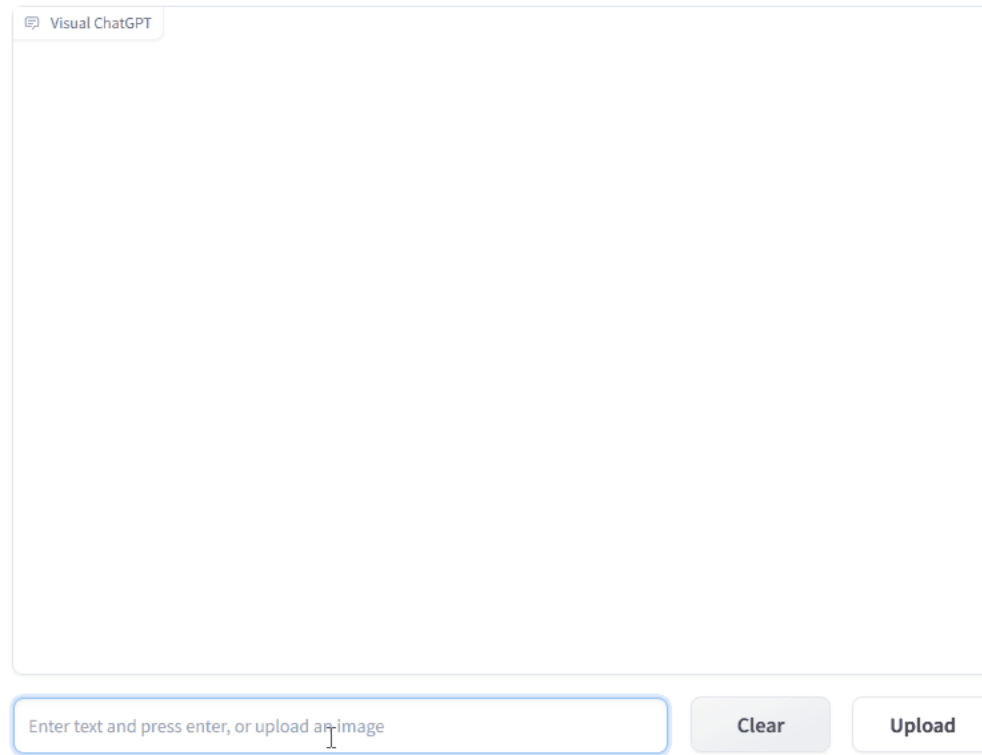


Image Generative 2022



Visual ChatGPT

Enter text and press enter, or upload an image

Clear Upload

The image shows a web interface for 'Visual ChatGPT'. It features a large, empty rectangular area for chat messages. At the bottom, there is a text input field with a light blue border and a small cursor. To the right of the input field are two buttons: 'Clear' and 'Upload'. The 'Clear' button is grey, and the 'Upload' button is white with a blue border. The overall design is clean and modern.

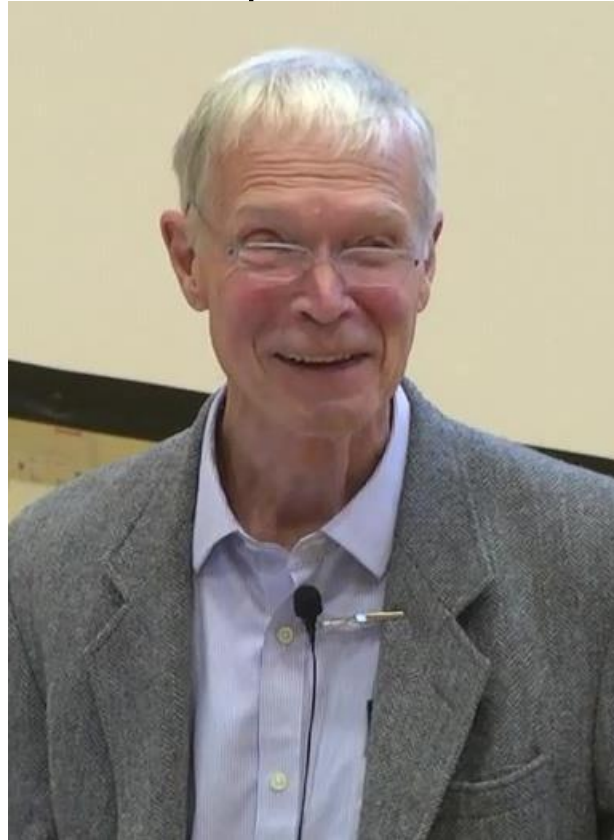
Nobel Prize in Physics 2024

Hinton speaking at the Nobel Prize
Lectures in Stockholm in 2024



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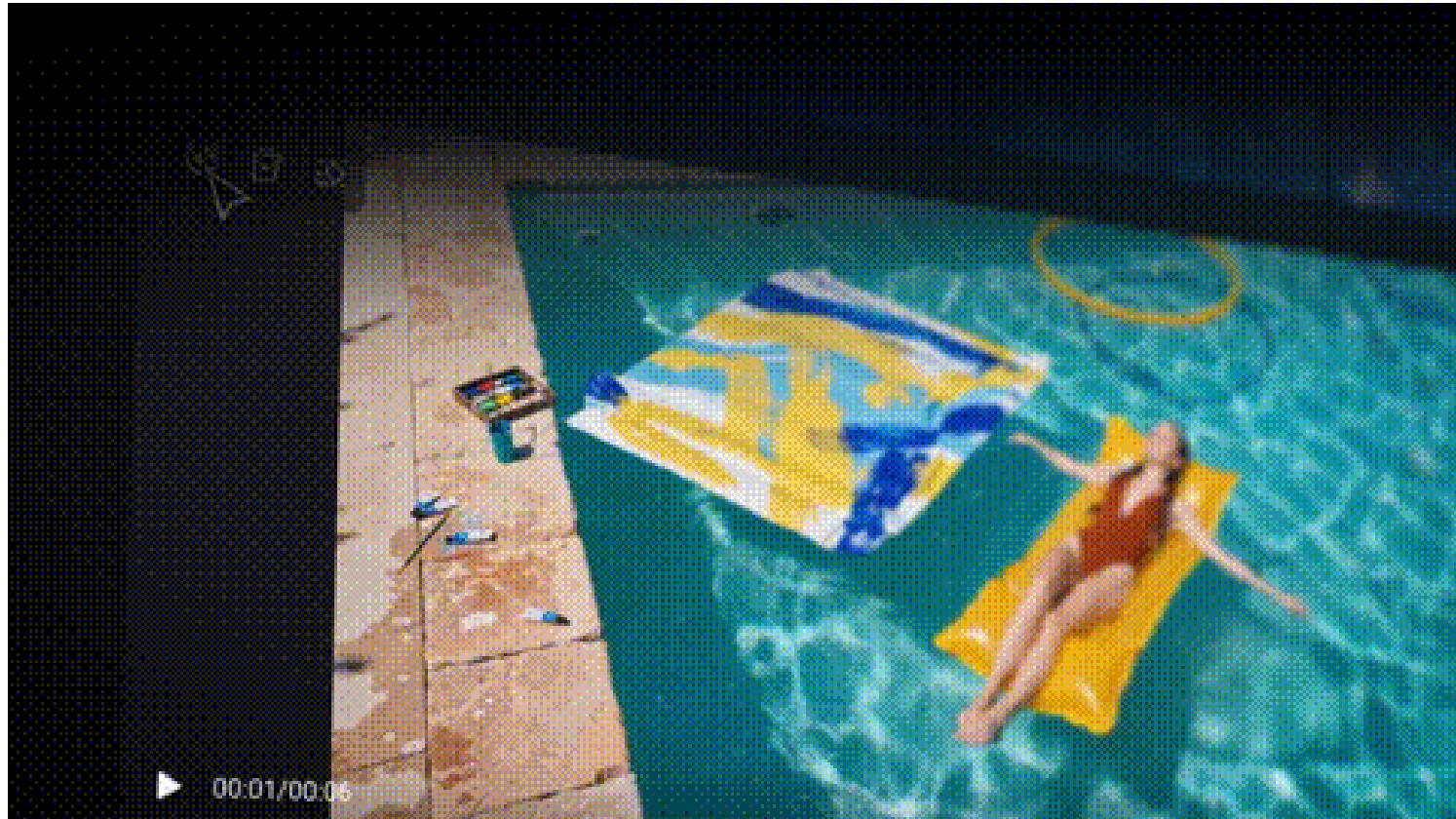
John Hopfield in 2016



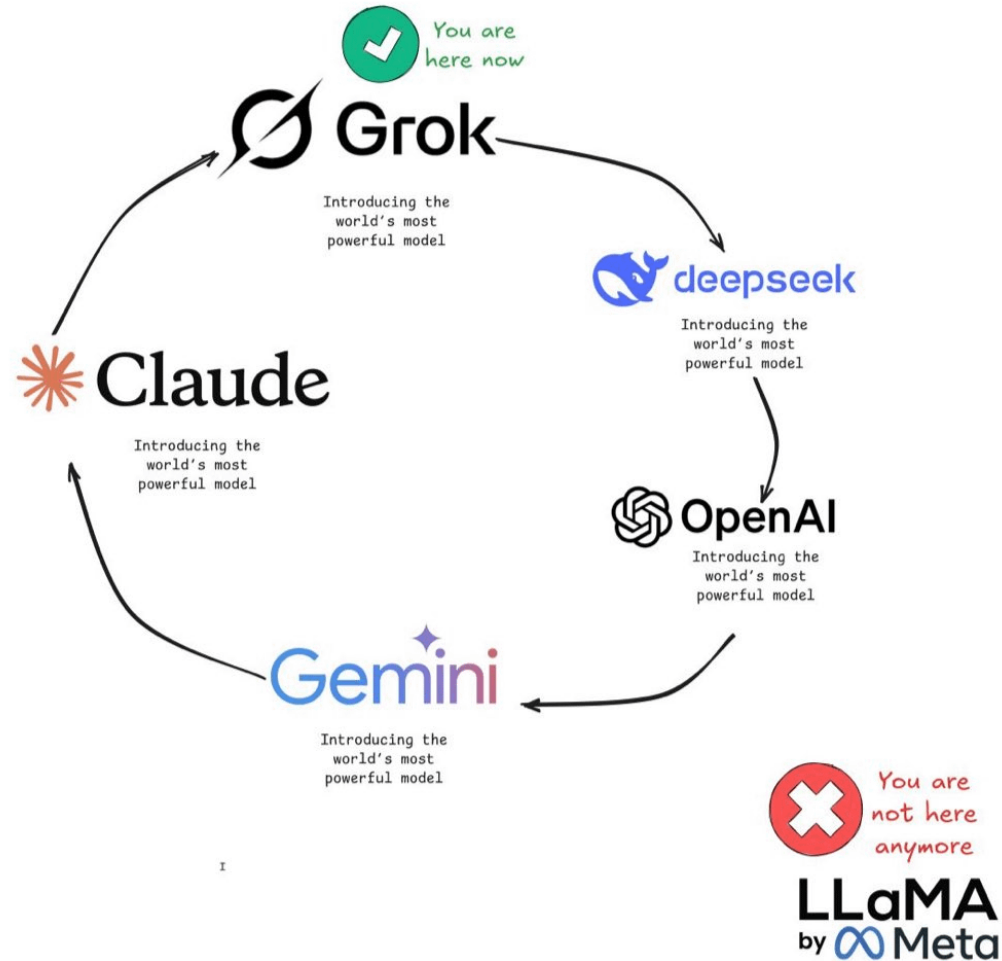
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In 2024, he was jointly awarded the [Nobel Prize in Physics](#) with [John Hopfield](#) “for foundational discoveries and inventions that enable machine learning with artificial neural networks.”

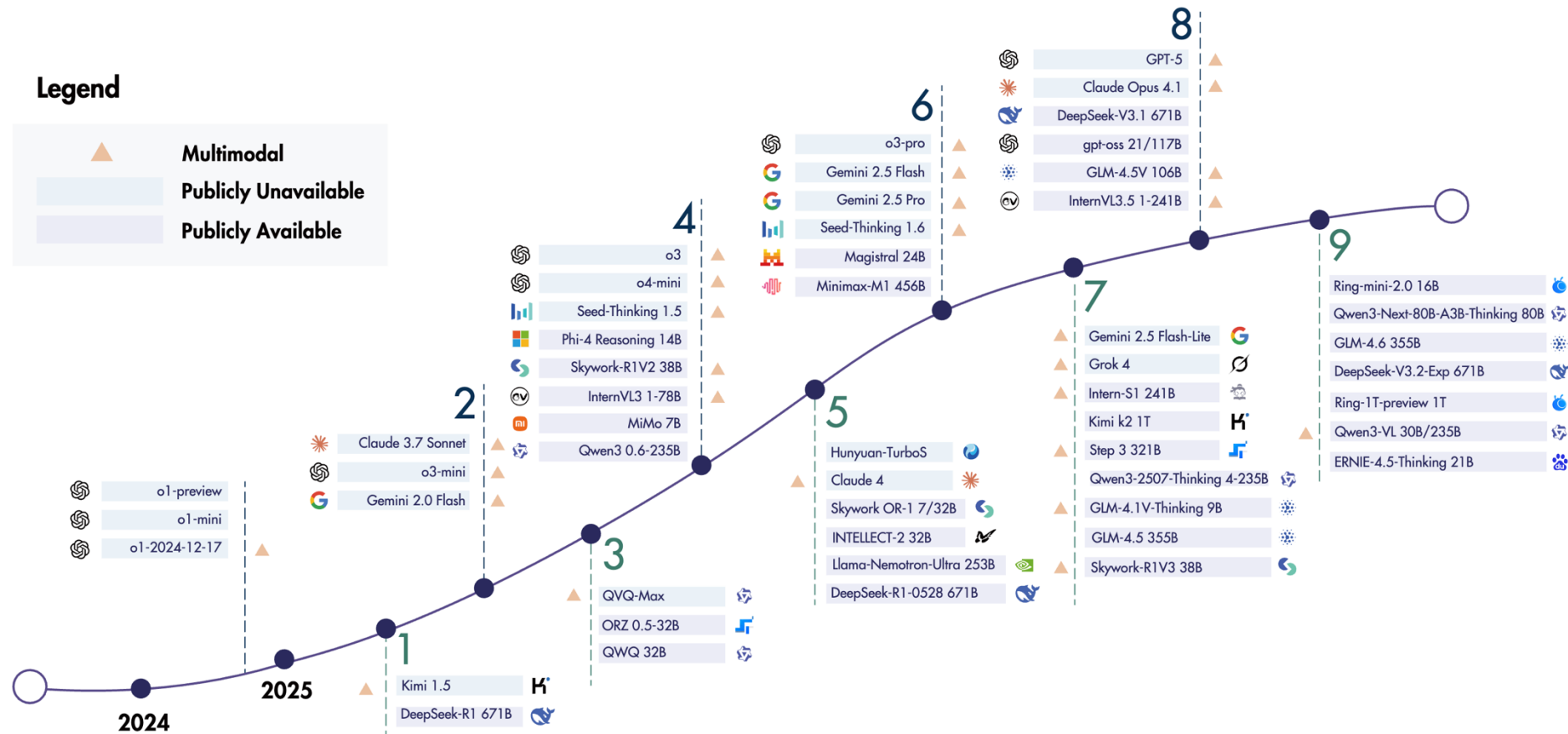
Video Generation 2025



Chat models: Improvement continues



Large Reasoning Models 2025



The Course

- Assignments with Google Collab (we'll give credits; 25%)
- Hands-on tutorials
- Lectures:
 - DL for CV (cs231n)
 - Years 2014-2019
 - Multimodal
 - Generative Image/Video
 - LLMs/LRMs (cs336)
 - Years 2023-2025